

(A never-ending) collection of notes on positive scalar curvature*

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September 2026

This is a growing collection of reading notes on positive scalar curvature. I may or may not have the ability to keep reading.

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*Last update: Sep 22, 2026.

1 Spin and scalar curvature in the presence of a fundamental group. I

1.1 Torus positive scalar curvature

Manifolds are smooth and without boundary.

Theorem 1.1 ([GL80, Theorem A]). *Let X be an enlargeable manifold in the sense of Gromov–Lawson. X carries no Riemannian metric of positive scalar curvature.*

Corollary 1.2 ([GL80, Corollary A]). *Any metric of nonnegative scalar curvature on T^n is flat.*

$X = X_0 \times S^2$, the product of any compact manifold X_0 with S^2 , carries a metric of positive scalar curvature no matter how large $\pi_1(X_0 \times S^2) = \pi_1(X_0)$ is [GL80].

Let X and Y be Riemannian manifolds with metrics g_X and g_Y . For $\varepsilon > 0$, the metric $g = \varepsilon g_X + g_Y$ on $X \times Y$ has scalar curvature $\kappa = \varepsilon^{-1} \kappa_X + \kappa_Y$.

Take $X = S^2(1)$, $Y = X_0$, and $g = \varepsilon g_{S^2(1)} + g_{X_0}$. Since $\kappa_{S^n(1)} = n(n-1)$, setting $n = 2$ gives $\kappa_{S^2(1)} = 2$, and therefore $\kappa = \frac{2}{\varepsilon} + \kappa_{X_0}$. If X_0 is compact, then $\kappa_{X_0} \geq -C \Rightarrow \kappa \geq \frac{2}{\varepsilon} - C$. Hence $\kappa > 0$ for sufficiently small ε .

If $\varepsilon = r^2$, $\kappa_{S^2(r)} = \frac{2}{r^2}$. So $\kappa_{X_0 \times S^2(r)} = \kappa_{X_0} + \frac{2}{r^2}$.

Let $p : \tilde{X} \rightarrow X$ be a covering map, and let g be a metric on X . Set $\tilde{g} = p^*g$. Then $\tilde{g}_{\tilde{x}}(V, W) = g_{p(\tilde{x})}(p_*V, p_*W)$. Hence $p : (\tilde{X}, \tilde{g}) \rightarrow (X, g)$ is a local isometry, so $\kappa_{\tilde{g}}(\tilde{x}) = \kappa_g(p(\tilde{x}))$. Thus $\kappa_{\tilde{g}} \geq \kappa_0 \Rightarrow \kappa_{\tilde{g}} \geq \kappa_0$.

Definition 1.3 ([LM89, Definition IV.5.1]). Let $f : X \rightarrow Y$ be a smooth map between Riemannian manifolds. f is ε -contracting if $\|f_*V\| \leq \varepsilon\|V\|$ for every V tangent to X , where $f_* : T_xX \rightarrow T_{f(x)}Y$.

For every piecewise smooth curve γ in X , $\text{length}(f(\gamma)) \leq \varepsilon \text{length}(\gamma)$ [LM89].

Note 1.4 ([LM89, (IV.5.1)]). For $f : X \rightarrow S^n$,

$$\deg(f) = \frac{\int_X f^* \omega}{\int_{S^n} \omega},$$

where ω is an n -form on S^n with nonzero integral. Here X and S^n are compact oriented manifolds of the same dimension.

The existence of an ε -contracting map $f : X \rightarrow Y$ of nonzero degree means that, in a strong sense, X is bigger than Y by a factor of at least $\frac{1}{\varepsilon}$ [LM89].

A map from a noncompact manifold is constant at infinity if it is constant outside a compact subset. For a map $f : X \rightarrow S^n$ from an oriented n -manifold that is constant at infinity, the formula in Note 1.4 defines $\deg(f)$ [LM89].

Definition 1.5 ([LM89, Definition IV.5.2]). A compact Riemannian n -fold is enlargeable if, for every $\varepsilon > 0$, there is an orientable Riemannian covering $\tilde{X} \rightarrow X$ that admits an ε -contracting map $f : \tilde{X} \rightarrow S^n(1)$ that is onto, constant at infinity, and of nonzero degree.

If the covering can be chosen finite for every $\varepsilon > 0$, then X is compactly enlargeable [LM89].

$\mathbb{R}^n/(k\mathbb{Z})^n \rightarrow \mathbb{R}^n/\mathbb{Z}^n$ is a k^n -fold covering, with $\mathbb{R}^n/(k\mathbb{Z})^n = \tilde{T}^n$ and $\mathbb{R}^n/\mathbb{Z}^n = T^n$. With the standard flat metrics, the map $[x] \mapsto [x/k]$ is an orientation-preserving diffeomorphism $\tilde{T}^n \rightarrow T^n$ and is $\frac{1}{k}$ -contracting. Composing it with a fixed smooth map $T^n \rightarrow S^n(1)$ of degree 1, whose differential is bounded by compactness, gives a map $f : \tilde{T}^n \rightarrow S^n(1)$ of degree 1 that is ε -contracting for sufficiently large k . So T^n is compactly enlargeable, and this property is independent of the Riemannian metric [LM89, proof of Theorem IV.5.3].

1.2 Clifford algebra

Let V be a vector space over a commutative field k , and let q be a quadratic form on V . Its tensor algebra is $\mathcal{T}(V) = \bigoplus_{r=0}^{\infty} \bigotimes^r V$. The ideal $I_q(V)$ in $\mathcal{T}(V)$ is generated by $v \otimes v + q(v)1$, $v \in V$ [LM89, Chapter I, §1].

$\text{Cl}(V, q) = \mathcal{T}(V)/I_q(V)$. Its defining relation is $v \cdot v = -q(v)1$. Assume $\text{char}(k) \neq 2$. Then $v \cdot w + w \cdot v = -2q(v, w)$, where $2q(v, w) = q(v + w) - q(v) - q(w)$. In the positive-definite case, if $e_1, \dots, e_n$ is a q -orthonormal basis, then $e_j^2 = -1$ and $e_j e_k = -e_k e_j$ for $j \neq k$ [LM89, (I.1.3)–(I.1.4)].

$$\text{Cl}^\times(V, q) = \{\varphi \in \text{Cl}(V, q) : \exists \varphi^{-1}, \varphi^{-1}\varphi = \varphi\varphi^{-1} = 1\} \text{ [LM89, (I.2.1)].}$$

Definition 1.7 ([LM89, (I.1.8); Definition I.2.3]). The Pin group $\text{Pin}(V, q)$ is the subgroup of $\text{Cl}^\times(V, q)$ generated by $v \in V$ with $q(v) = \pm 1$.

$\text{Spin}(V, q) = \text{Pin}(V, q) \cap \text{Cl}^0(V, q)$. Here $\text{Cl}^0(V, q)$ denotes the even part of $\text{Cl}(V, q)$.

For the standard positive-definite form on $\mathbb{R}^n$, write $\text{Spin}_n = \text{Spin}(\mathbb{R}^n, \|\cdot\|^2)$ [LM89, (I.2.30)–(I.2.32)].

Let $\pi : E \longrightarrow X$ be a real oriented n -dimensional vector bundle over X , equipped with a Riemannian structure. $P_{\text{SO}_n}(E)$ is the bundle of oriented orthonormal frames. Assume $n \geq 3$.

Definition 1.8 ([LM89, Definition II.1.3 and the diagram of fibrations]). A spin structure on E is a principal Spin_n -bundle $P_{\mathrm{Spin}_n}(E)$ together with a 2-sheeted covering $\xi : P_{\mathrm{Spin}_n}(E) \longrightarrow P_{\mathrm{SO}_n}(E)$ such that $\xi(pg) = \xi(p)\xi_0(g)$ for all $p \in P_{\mathrm{Spin}_n}(E)$ and $g \in \mathrm{Spin}_n$.

$$\begin{array}{ccccc}
& & \text{Spin}_n & \xrightarrow{\xi_0} & \text{SO}_n \\
\mathbb{Z}_2 \swarrow & & \downarrow & & \downarrow \\
& & P\text{Spin}_n(E) & \xrightarrow{\xi} & P\text{SO}_n(E) \\
& & \searrow \pi' & & \swarrow \pi \\
& & X & &
\end{array}$$

ξ_0 is the connected 2-fold covering, and ξ is a 2-sheeted covering.

For $n = 2$, the definition is analogous, using the connected 2-fold covering $\mathrm{SO}_2 \longrightarrow \mathrm{SO}_2$; for $n = 1$, a spin structure is a 2-fold covering of $X = P_{\mathrm{SO}_1}(E)$.

Theorem 1.9 ([LM89, Theorem II.1.7]). *There exists a spin structure on E if and only if the second Stiefel–Whitney class of E is zero. If $w_2(E) = 0$, the distinct spin structures on E are in one-to-one correspondence with the elements of $H^1(X; \mathbb{Z}_2)$.*

A spin manifold is an oriented Riemannian manifold with a spin structure on its tangent bundle [LM89, Chapter II, §2]. For the torus, $TT^n \cong T^n \times \mathbb{R}^n$ is trivial, so all its positive-degree Stiefel–Whitney classes vanish; in particular, $w_2(TT^n) = 0$. By Theorem 1.9, T^n is spin. Similarly, $\mathbb{R}^n/(k\mathbb{Z})^n$ is spin. Thus T^n is enlargeable.

1.3 Complex spinor bundle

Let X be a compact Riemannian spin manifold of dimension $2n$. The Clifford algebra bundle $\text{Cl}(X)$ has fibre $\text{Cl}(T_x X)$ at x . $\mathcal{S}_{\mathbb{C}}(X) = P_{\text{Spin}_{2n}}(X) \times_{\mu_{\mathbb{C}}} M_{\mathbb{C}}$, where $M_{\mathbb{C}} \cong \mathbb{C}^{2^n}$ is the irreducible complex module over $\text{Cl}(\mathbb{R}^{2n})$ and $\mu_{\mathbb{C}} : \text{Spin}_{2n} \rightarrow \text{GL}_{\mathbb{C}}(M_{\mathbb{C}})$ is the restriction of its Clifford action to Spin_{2n} [GL80].

The Riemannian connection on $P_{\text{SO}_{2n}}(X)$ lifts through $P_{\text{Spin}_{2n}}(X) \rightarrow P_{\text{SO}_{2n}}(X)$ and induces a connection on $\mathcal{S}_{\mathbb{C}}(X)$. $\mathcal{S}_{\mathbb{C}}(X)$ is a complex Dirac bundle [LM89, discussion following Theorem II.4.17; Example II.5.9]. This means that unit tangent vectors act unitarily and the metric connection satisfies $\nabla_V(\varphi \cdot \sigma) = (\nabla_V \varphi) \cdot \sigma + \varphi \cdot \nabla_V \sigma$ for sections φ of $\text{Cl}(X)$ and σ of $\mathcal{S}_{\mathbb{C}}(X)$ [LM89, Definition II.5.2].

1.3.1 Some results

- $D\sigma = \sum_{j=1}^{2n} e_j \cdot \nabla_{e_j} \sigma$, where $e_1, \dots, e_{2n}$ are local orthonormal vector fields. This is the Atiyah–Singer operator [LM89, (II.5.0); Example II.5.9].
- For compactly supported sections σ, τ , $\langle D\sigma, \tau \rangle_{L^2} = \langle \sigma, D\tau \rangle_{L^2}$ [LM89, Proposition II.5.3].
- D is elliptic [LM89, Lemma II.5.1].

Let S be a Dirac bundle over X with connection ∇ . $\nabla_{V,W}^2 \sigma = \nabla_V \nabla_W \sigma - \nabla_{\nabla_V W} \sigma$. $\nabla_{V,W}^2 - \nabla_{W,V}^2 = R_{V,W}$ [GL80].

$$\nabla^* \nabla \sigma = -\text{trace}(\nabla^2 \sigma) = -\sum_{j=1}^{2n} \nabla_{e_j, e_j}^2 \sigma.$$

$\langle \nabla^* \nabla \sigma, \tau \rangle_{L^2} = \langle \nabla \sigma, \nabla \tau \rangle_{L^2}$. $\langle \nabla^* \nabla \sigma, \sigma \rangle_{L^2} = \|\nabla \sigma\|_{L^2}^2 \geq 0$; if this is zero, then $\nabla \sigma = 0$. So $\ker(\nabla^* \nabla) = \{\text{parallel sections}\}$ [GL80].

1.3.2 Bochner formula

Define the curvature endomorphism $\mathcal{R} : S \rightarrow S$ by $\mathcal{R}(\sigma) = \frac{1}{2} \sum_{j,k=1}^{2n} e_j \cdot e_k \cdot R_{e_j, e_k}(\sigma)$, where $R_{V,W} : S_x \rightarrow S_x$.

Theorem 1.10 ([GL80, Theorem 1.1 and (1.1)]).

$$D^2 = \nabla^* \nabla + \mathcal{R}. \quad (1.1)$$

Apply Theorem 1.10 to $\mathcal{S}_{\mathbb{C}}(X)$.

$$R_{V,W}^S(\sigma) = \frac{1}{2} \sum_{a < b} \langle R_{V,W}(e_a), e_b \rangle e_a \cdot e_b \cdot \sigma,$$

where $R_{V,W}$ is the Riemann curvature tensor of X , and the products denote Clifford multiplication. Then $\mathcal{R} = \frac{1}{4}\kappa$, and (1.1) becomes

$$D^2 = \nabla^* \nabla + \frac{1}{4}\kappa. \quad (1.2)$$

This is the Lichnerowicz formula. Thus $\kappa > 0 \Rightarrow \ker(D) = 0$ [GL80].

Now let $E \rightarrow X$ be a Hermitian vector bundle equipped with a unitary connection ∇^E . $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ is a bundle of modules over $\text{Cl}(X)$ and is also a complex Dirac bundle. $\nabla_V(\sigma \otimes e) = (\nabla_V^S \sigma) \otimes e + \sigma \otimes \nabla_V^E e$ [LM89, Proposition II.5.10; note preceding Theorem II.8.17].

D_E is the Dirac operator of the twisted spinor bundle. $D_E(\sigma \otimes e) = \sum_{j=1}^{2n} e_j \cdot \nabla_{e_j}(\sigma \otimes e)$.

$$R_{e_j, e_k}^{S \otimes E} = R_{e_j, e_k}^S \otimes 1 + 1 \otimes R_{e_j, e_k}^E \cdot \frac{1}{2} \sum_{j,k} e_j \cdot e_k \cdot (R_{e_j, e_k}^S \otimes 1) = \frac{1}{4}\kappa.$$

Following [GL80, (1.3)], define $\mathcal{R}^E : \mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E \rightarrow \mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ by

$$\mathcal{R}^E(\sigma \otimes e) = \frac{1}{2} \sum_{j,k=1}^{2n} (e_j \cdot e_k \cdot \sigma) \otimes R_{e_j, e_k}^E(e). \quad (1.3)$$

Theorem 1.11 ([LM89, Theorem II.8.17 and (II.8.23)]).

$$D_E^2 = \nabla^* \nabla + \frac{1}{4}\kappa + \mathcal{R}^E. \quad (1.4)$$

Below, endomorphism norms are the suprema of the fibrewise operator norms; curvature two-form norms also take the supremum over unit tangent vectors in both slots.

Theorem 1.12 ([GL80, Theorem 1.3]; compare [LM89, (IV.5.5)]). *Let X be a compact Riemannian spin manifold, and let $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$ be a twisted spinor bundle. If $\kappa \geq \kappa_0 > 0$ and $\|\mathcal{R}^E\| < \frac{\kappa_0}{4}$, then $\ker D_E = 0$.*

From $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E$, one can construct a natural elliptic complex.

Let $e_1, \dots, e_{2n}$ be local positively oriented orthonormal vector fields. $\omega_{\mathbb{C}} = i^n e_1 \cdots e_{2n}$ is a parallel section of $\text{Cl}(X) \otimes_{\mathbb{R}} \mathbb{C}$, so $\nabla \omega_{\mathbb{C}} = 0$. By the Clifford relations, $\omega_{\mathbb{C}}^2 = 1$ and $e_j \omega_{\mathbb{C}} = -\omega_{\mathbb{C}} e_j$ for every j .

$$\mathcal{S}_{\mathbb{C}}(X) = \underbrace{\mathcal{S}_{\mathbb{C}}^+(X)}_{\{\sigma : \omega_{\mathbb{C}} \cdot \sigma = \sigma\}} \oplus \underbrace{\mathcal{S}_{\mathbb{C}}^-(X)}_{\{\sigma : \omega_{\mathbb{C}} \cdot \sigma = -\sigma\}}. \quad (1.5)$$

Tensoring (1.5) with E gives $\mathcal{S}_{\mathbb{C}}(X) \otimes_{\mathbb{C}} E = (\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E) \oplus (\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E)$. Since $\omega_{\mathbb{C}}$ is parallel and anticommutes with tangent vectors, D_E interchanges the two summands. With respect to this splitting,

$$D_E = \begin{pmatrix} 0 & D_E^- \\ D_E^+ & 0 \end{pmatrix}. \quad (1.6)$$

Here $D_E^- : \Gamma(\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E) \rightarrow \Gamma(\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E)$.

The twisted Atiyah–Singer operator is $D_E^+ : \Gamma(\mathcal{S}_{\mathbb{C}}^+(X) \otimes_{\mathbb{C}} E) \rightarrow \Gamma(\mathcal{S}_{\mathbb{C}}^-(X) \otimes_{\mathbb{C}} E)$. D_E is formally self-adjoint. $D_E^- = (D_E^+)^*$ [GL80, (1.4)].

1.3.3 Analytic index

If X is compact and D_E^+ is elliptic, then its kernel and cokernel are finite-dimensional. Its analytic index is

$$\text{ind}(D_E^+) = \dim_{\mathbb{C}} \ker D_E^+ - \dim_{\mathbb{C}} \text{coker } D_E^+. \quad (1.7)$$

Ellipticity on a compact manifold implies that the range is closed. So $\text{coker } D_E^+ \cong \ker((D_E^+)^*) \cong \ker(D_E^-)$. By (1.6), $\ker(D_E) = \ker D_E^+ \oplus \ker D_E^-$ [LM89, Theorem III.5.2(ii) and (III.5.2)].

By Theorem 1.12, $\ker D_E = 0 \Rightarrow \ker D_E^+ = 0$, $\ker D_E^- = 0$, and $\text{coker } D_E^+ = 0$. So (1.7) gives $\text{ind}(D_E^+) = 0$. When $\kappa > 0$, take E to be the trivial flat complex line bundle. Then $R^E = 0$, and the same argument using (1.2) gives $\text{ind}(D^+) = 0$ [GL80, Theorem 1.4].

1.3.4 Topological index

Let E be a complex vector bundle of rank d . Its total Chern class is $c(E) = 1 + c_1(E) + \cdots + c_d(E)$, where, formally, $c(E) = \prod_{k=1}^d (1 + x_k)$.

The Chern character is

$$\text{ch}(E) = e^{x_1} + \cdots + e^{x_d} = d + \sum_{j=1}^d x_j + \frac{1}{2} \sum_{j=1}^d x_j^2 + \cdots \in H^{2*}(X; \mathbb{Q}) \quad [\text{LM89, (III.11.22)}].$$

$\text{ch}^k(E) = \frac{1}{k!} \sum_{j=1}^d x_j^k$. $\text{ch}(E \oplus E') = \text{ch}(E) + \text{ch}(E')$. $\text{ch}(E \otimes E') = \text{ch}(E) \text{ch}(E')$ [LM89, (III.11.23) and Proposition III.11.16].

$H^{2j}(S^{2n}; \mathbb{Z}) = 0$ for $0 < j < n$. Let $E_0 \rightarrow S^{2n}(1)$ have rank d . Then $c_j(E_0) = 0$ for $0 < j < n$. If $c_n(E_0) \neq 0$, the formulas [LM89, (III.11.22)–(III.11.23)] give

$$\text{ch}(E_0) = d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E_0). \quad (1.8)$$

Example 1.13 ([LM89, Example III.11.13 and (III.11.17)]). The $\widehat{A}$ -sequence is determined by $\widehat{a}(x) = \frac{\sqrt{x}/2}{\sinh(\sqrt{x}/2)}$. For a real oriented $2m$ -dimensional vector bundle E , write formally $E \otimes_{\mathbb{R}} \mathbb{C} = \ell_1 \oplus \bar{\ell}_1 \oplus \cdots \oplus \ell_m \oplus \bar{\ell}_m$ according to the Splitting Principle and set $x_j = c_1(\ell_j)$. Then

$$\widehat{A}(E) = \prod_{j=1}^m \frac{x_j/2}{\sinh(x_j/2)}, \quad \widehat{A}(E) = 1 + \widehat{A}_1(p_1 E) + \widehat{A}_2(p_1 E, p_2 E) + \cdots.$$

$\widehat{A}(TX) \in H^{4*}(X; \mathbb{Q})$. $\widehat{A}(X) = \{\widehat{A}(TX)\}[X]$, where the top-degree part is evaluated on the fundamental class $[X]$. So $\widehat{A}(X) = 0$ whenever $\dim X \not\equiv 0 \pmod{4}$ [LM89, (III.11.19)].

Recall that $\text{ind}(D_E^+) = \dim_{\mathbb{C}} \ker D_E^+ - \dim_{\mathbb{C}} \text{coker } D_E^+$.

Theorem 1.14 ([LM89, Theorem III.13.10 and (III.13.26)]). *For the twisted Atiyah–Singer operator,*

$$\text{ind}(D_E^+) = \{\text{ch}(E) \widehat{A}(TX)\}[X]. \quad (1.9)$$

For the trivial complex line bundle E , $\text{ch}(E) = 1$, so Theorem 1.14 gives $\underline{\text{ind}}(D^+) = \widehat{A}(X)$.

1.4 Compactly enlargeable spin manifolds

If X is compactly enlargeable, then $X \times S^1$ is compactly enlargeable. If X is spin, then $X \times S^1$ is spin. If X carries a metric with $\kappa > 0$, then $X \times S^1$ carries a metric with $\kappa > 0$ [LM89, Proposition II.2.15; (IV.5.2)–(IV.5.3)].

Let X be a compactly enlargeable spin manifold, and suppose that X carries a Riemannian metric with $\kappa > 0$. Since X is compact, $\kappa \geq \kappa_0 > 0$ for some κ_0 . Replacing X by $X \times S^1$ if necessary, assume that X has dimension $2n$.

Choose a complex vector bundle $E_0 \rightarrow S^{2n}(1)$ of rank d such that $c_n(E_0) \neq 0$. Such a bundle exists by the Chern-character isomorphism for the even-dimensional sphere [LM89, proof of Theorem IV.5.5]. Fix a Hermitian metric and a unitary connection ∇^{E_0} on E_0 . Let R^{E_0} be its curvature 2-form. Its norm is finite because the sphere is compact.

Let $\varepsilon > 0$. Since X is compactly enlargeable, there is a finite orientable Riemannian covering $p : \widetilde{X} \rightarrow X$ that admits an ε -contracting map $f : \widetilde{X} \rightarrow S^{2n}(1)$ of nonzero degree. The metric on $\widetilde{X}$ is the lifted metric p^*g . Since p is a local isometry, $\kappa_{\widetilde{X}} = \kappa_X \circ p$, so $\kappa_{\widetilde{X}} \geq \kappa_0$ [LM89, Proof of Theorem IV.5.5].

Because X is spin, the pullback spin structure makes $\widetilde{X}$ spin; equivalently, $w_2(T\widetilde{X}) = p^*w_2(TX) = 0$. Thus the twisted Atiyah–Singer operator can be constructed on $\widetilde{X}$.

Let $E = f^*E_0$, $\nabla^E = f^*\nabla^{E_0}$. Under the identification $E_x = (E_0)_{f(x)}$, $R_{V,W}^E = R_{f_*V, f_*W}^{E_0}$.

Consider $\mathcal{S}_{\mathbb{C}}(\widetilde{X}) \otimes_{\mathbb{C}} E$. By Theorem 1.11, its Bochner formula is (1.4), where $\mathcal{R}^E$ is given by (1.3) and depends universally and linearly on the components of R^E . Thus, by [LM89, (IV.5.4) and (IV.5.7)], there is a constant k_n depending only on the dimension such that

$$\|\mathcal{R}^E\| \leq k_n \|R^E\|. \quad (1.10)$$

Moreover, $\|\mathcal{R}^E\| \leq \varepsilon^2 \|R^{E_0}\|$. As in [LM89, (IV.5.8)], combining (1.10) with this estimate gives

$$\|\mathcal{R}^E\| \leq k_n \varepsilon^2 \|R^{E_0}\| < \frac{\kappa_0}{4} \quad \text{for sufficiently small } \varepsilon;$$

$$\text{equivalently, } 0 < \varepsilon < \sqrt{\frac{\kappa_0}{4k_n \|R^{E_0}\|}}.$$

Then $\|\mathcal{R}^E\| < \frac{\kappa_0}{4}$. Since $\kappa \geq \kappa_0$, for every nonzero σ in a fibre of the twisted spinor bundle,

$$\left\langle \left(\frac{1}{4} \kappa + \mathcal{R}^E \right) \sigma, \sigma \right\rangle \geq \left(\frac{\kappa_0}{4} - \|\mathcal{R}^E\| \right) |\sigma|^2 > 0.$$

Integrating and using (1.4) and $\nabla^* \nabla \geq 0$, we get $D_E^2 > 0$, so $\ker D_E = 0$. Hence $\ker D_E^+ = 0$ and $\ker D_E^- = 0$. Since $\text{coker } D_E^+ \cong \ker D_E^-$, $\text{ind}(D_E^+) = 0$ on the analytic side.

By the Atiyah–Singer formula (1.9) and (1.8), using $\text{ch}(E) = f^* \text{ch}(E_0) = d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E)$,

$$\begin{aligned}
 \text{ind}(D_E^+) &= \{\text{ch}(E) \widehat{A}(T\widetilde{X})\}[\widetilde{X}] \\
 &= \left\{ \left(d + \frac{(-1)^{n-1}}{(n-1)!} c_n(E) \right) \widehat{A}(T\widetilde{X}) \right\}[\widetilde{X}] \\
 &= d \underbrace{\widehat{A}(\widetilde{X})}_{\text{ind}(D^+) = 0} + \frac{(-1)^{n-1}}{(n-1)!} c_n(E)[\widetilde{X}] \\
 &= \frac{(-1)^{n-1}}{(n-1)!} c_n(E)[\widetilde{X}] \\
 &= \frac{(-1)^{n-1}}{(n-1)!} \underbrace{f^* c_n(E_0)[\widetilde{X}]}_{\substack{= \deg(f) c_n(E_0)[S^{2n}] \\ \deg(f) \neq 0}} \neq 0.
 \end{aligned}$$

See [LM89, (IV.5.9); (III.11.22)–(III.11.23)].

So a compactly enlargeable spin manifold cannot carry a metric with positive scalar curvature [GL80, Proposition 3.1; Theorem 3.7]; compare [LM89, Theorem IV.5.5].

1.5 The torus

By Example 1.6, T^n is compactly enlargeable; T^n and its finite covering tori are spin.

- (1) $n = 2m$. Suppose $\kappa > 0$, so $\kappa \geq \kappa_0 > 0$ for some κ_0 . Compact enlargeability gives a finite covering $p : \widetilde{T}^{2m} \rightarrow T^{2m}$ and an ε -contracting map $f : \widetilde{T}^{2m} \rightarrow S^{2m}(1)$ of nonzero degree for $\varepsilon > 0$ sufficiently small. Since $\kappa_{\widetilde{T}^{2m}} \geq \kappa_0$ and $\widetilde{T}^{2m}$ is spin, the preceding contradiction applies. Thus T^{2m} carries no metric with $\kappa > 0$.
- (2) $n = 2m + 1$. Suppose T^{2m+1} carries a metric g with $\kappa_g > 0$. $T^{2m+1} \times S^1 = T^{2m+2}$.

$$\kappa_{g+dt^2} = \kappa_g + \kappa_{S^1} = \kappa_g > 0,$$

which contradicts the even-dimensional case above [LM89, (IV.5.2)–(IV.5.3)].

Theorem 1.15 ([LM89, Theorem IV.5.7 (J. P. Bourguignon)]). *Let X be a compact connected manifold that carries no metric with $\kappa > 0$. Then every metric on it with $\kappa \geq 0$ is Ricci flat.*

Proposition 1.16 ([LM89, Proposition IV.5.8]). *A Ricci-flat enlargeable manifold is flat.*

Applying Theorem 1.15 and Proposition 1.16 to T^n gives Corollary 1.2.

2 A proof of Gromov's cube inequality on scalar curvature

2.1 Cube inequality

Theorem 2.1 ([WXY23, Theorem 1.1]). *Let g be a Riemannian metric on the cube $I^n = [0, 1]^n$, $n \geq 2$. Suppose that $\ell_i = \text{dist}(\partial_{i-}, \partial_{i+})$ is the g -distance between the i -th pair of opposite faces ∂_{i-} and ∂_{i+} of the cube. Then:*

(1) *If $\text{Sc}_g \geq k > 0$, then*

$$\sum_{i=1}^n \frac{1}{\ell_i^2} \geq \frac{kn}{4\pi^2(n-1)}.$$

Consequently,

$$\min_{1 \leq i \leq n} \text{dist}(\partial_{i-}, \partial_{i+}) \leq 2\pi \sqrt{\frac{n-1}{k}}.$$

(2) *If $\text{Sc}_g \geq k > 0$ and the dihedral angles of (I^n, g) are $< \pi$, then*

$$\sum_{i=1}^n \frac{1}{\ell_i^2} > \frac{kn}{4\pi^2(n-1)}.$$

Consequently,

$$\min_{1 \leq i \leq n} \text{dist}(\partial_{i-}, \partial_{i+}) < 2\pi \sqrt{\frac{n-1}{k}}.$$

Remark 2.2. A point of I^n is $x = (x_1, \dots, x_n)$, where $0 \leq x_i \leq 1$.

$$\partial_{i-} = \{x \in I^n : x_i = 0\} \quad \partial_{i+} = \{x \in I^n : x_i = 1\}.$$

For $I^2 = [0, 1]^2$,

$$\begin{aligned} \partial_{1-} &= \{0\} \times [0, 1], & \partial_{1+} &= \{1\} \times [0, 1], \\ \partial_{2-} &= [0, 1] \times \{0\}, & \partial_{2+} &= [0, 1] \times \{1\}. \end{aligned}$$

Proof. Part I: $\geq$ inequality

By rescaling the metric, it suffices to assume $k = n(n-1)$.

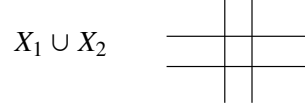
Step 1. Assume to the contrary that $\sum_{i=1}^n \frac{1}{\ell_i^2} < \frac{n^2}{4\pi^2}$. Consequently, $\sum_{i=1}^n \frac{\pi^2}{\ell_i^2} < \frac{n^2}{4}$.

As $\varepsilon \rightarrow 0$, $\frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} \rightarrow \frac{\pi^2}{\ell_i^2}$. This depends continuously on ε . Hence there exists $\varepsilon > 0$ such that $0 < 2\varepsilon < \min_i \ell_i$, $\varepsilon < \min_i \pi^2 / \ell_i^2$, and

$$\sum_{i=1}^n \frac{\pi^2 (1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} < \frac{n^2}{4}.$$

Step 2. Extend the metric on I^n to a complete metric on $\mathbb{R}^n$.

(a) The case $n = 2$. Let $X = [0, 1]^2$, $X_1 = [0, 1] \times \mathbb{R}$, and $X_2 = \mathbb{R} \times [0, 1]$. Then $X_1 \cap X_2 = X$.



Original metric: $g = \sum_{a,b=1}^2 g_{ab} dx_a dx_b$.

g on $[0, 1]^2$:

$$\ell_1 = \text{dist}_g(\{0\} \times [0, 1], \{1\} \times [0, 1]),$$

$$\ell_2 = \text{dist}_g([0, 1] \times \{0\}, [0, 1] \times \{1\}).$$

Whitney's extension theorem and a smooth partition of unity extend the smooth coefficient functions, preserve a positive-definite symmetric form, and transition to the Euclidean metric. Thus $g_1 = g$ on X , while $g_1 = dx_1^2 + dx_2^2$ outside a compact set of $X_1 \cup X_2$.

$$\begin{aligned} \partial_- X_1 &= \{0\} \times \mathbb{R} & \partial_+ X_1 &= \{1\} \times \mathbb{R} \\ \partial_- X_2 &= \mathbb{R} \times \{0\} & \partial_+ X_2 &= \mathbb{R} \times \{1\}, \end{aligned}$$

Difficulty: g_1 may create short curves outside X .

Choose a smooth positive function $\bar{\Psi}$ such that $\bar{\Psi} = 1$ on X , and $\bar{\Psi}$ is constant when $x_1^2 + x_2^2$ is sufficiently large. $g_2 := \bar{\Psi} g_1$.

$$L_{g_2}(\gamma) = \int \sqrt{\bar{\Psi}(\gamma(t))} |\gamma'(t)|_{g_1} dt.$$

Choose $\bar{\Psi}$ large enough so that, on $X_1 \cup X_2$, g_2 satisfies

$$(1) \quad \text{dist}_{g_2}(\partial_- X_1, \partial_+ X_1) \geq \ell_1 - \frac{\varepsilon}{3},$$

$$(2) \quad \text{dist}_{g_2}(\partial_- X_2, \partial_+ X_2) \geq \ell_2 - \frac{\varepsilon}{3}$$

$$g_2 = \underbrace{R^2}_{\bar{\Psi} \text{ constant}} (dx_1^2 + dx_2^2) \quad \text{outside of a compact subset of } X_1 \cup X_2.$$

Extend g_2 to a **complete** metric $\bar{g}$ on $\mathbb{R}^2$ such that

$$\bar{g} = \underbrace{R^2}_{\text{constant}} \underbrace{(dx_1^2 + dx_2^2)}_{\text{Euclidean}} \quad \text{outside a compact subset of } \mathbb{R}^2.$$

Thus $\bar{g}$ is flat and $\text{Sc}_{\bar{g}} = 0$ outside a compact set. On the remaining compact set, $\text{Sc}_{\bar{g}}$ is continuous, so it has a finite minimum. **Continuity $\Rightarrow$ finite minimum**. Choose $\sigma > 0$ sufficiently large so that $\text{Sc}_{\bar{g}} \geq -\sigma$ on $\mathbb{R}^2$.

(b) *General construction.*

For $k = 0, 1, \dots, n$, Λ_k is the family of all subsets of $\{1, \dots, n\}$ with k elements. For $\lambda \in \Lambda_k$, set $X_\lambda = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n : 0 \leq x_j \leq 1 \text{ if } j \notin \lambda\}$. Thus the coordinates indexed by λ are unrestricted, while the other coordinates lie in $[0, 1]$. $X_\emptyset = [0, 1]^n$.

For $j \notin \lambda$, the faces on $x_j = 0$ and $x_j = 1$ are $\partial_{j-}X_\lambda$ and $\partial_{j+}X_\lambda$, respectively.

$$\partial_{j-}X_\lambda = \{x_j = 0\} \cap X_\lambda, \quad \partial_{j+}X_\lambda = \{x_j = 1\} \cap X_\lambda.$$

We construct the metric by induction so that, for $0 \leq k \leq n-1$, $\lambda \in \Lambda_k$, and $j \notin \lambda$,

$$\text{dist}_{\bar{g}}(\partial_{j-}X_\lambda, \partial_{j+}X_\lambda) \geq \ell_j - \frac{k\varepsilon}{n+1} \quad (2.3)$$

These distances are intrinsic in X_λ ; the case $k = 0$ holds by definition. At each of the k stages, each remaining face distance decreases by at most $\frac{\varepsilon}{n+1}$, giving $\frac{k\varepsilon}{n+1}$ after k stages.

Induction step: k to $k+1$.

Assume the metric has been constructed on the union of X_λ , $\lambda \in \Lambda_k$, satisfying (2.3). Let $\mu \in \Lambda_{k+1}$. The $k+1$ sets X_λ , where $\lambda \subset \mu$ and $|\lambda| = k$, are already contained in X_μ :

$$\bigcup_{\substack{\lambda \subset \mu \\ |\lambda|=k}} X_\lambda \subset X_\mu.$$

When two such sets overlap, their intersection is an X_λ from an earlier stage. Extend the metric from this union to X_μ using Whitney's extension theorem, a smooth partition of unity, and rescaling; an additional coordinate ranges over $\mathbb{R}$. At this stage, the loss changes from $k\varepsilon/(n+1)$ to $(k+1)\varepsilon/(n+1)$. After the final extension, $\bar{g} = g$ on I^n and $\bar{g} = R^2g_o$ outside a compact set, where g_o is the standard Euclidean metric and $R > 0$. Thus $\bar{g}$ is complete; choose $\sigma > 0$ so that $\text{Sc}_{\bar{g}} \geq -\sigma$.

Let $E_i = \{(y_1, \dots, y_n) \in \mathbb{R}^n : y_i = 0\}$.

$$\text{1-Lipschitz} \longrightarrow \varphi_i = \text{sgn}(x_i) \text{dist}_{\bar{g}}(x, E_i) = \begin{cases} \text{dist}_{\bar{g}}(x, E_i) & x_i > 0 \\ 0 & x_i = 0 \\ -\text{dist}_{\bar{g}}(x, E_i) & x_i < 0 \end{cases}$$

1-Lipschitz: $|\text{dist}_{\bar{g}}(x, E_i) - \text{dist}_{\bar{g}}(y, E_i)| \leq \text{dist}_{\bar{g}}(x, y)$. For points on opposite sides, every connecting path crosses E_i , so the sum of their distances to E_i is at most their mutual distance. Hence the signed function φ_i is also 1-Lipschitz.

$$\underbrace{\psi_i}_{\text{Centered.}} = \varphi_i - \frac{\ell_i}{2}, \quad x_i = 0 \Rightarrow \psi_i(x) = -\ell_i/2.$$

By smooth Lipschitz approximation, choose a smooth z_i such that

$$|z_i - \underbrace{\psi_i}_{1\text{-Lipschitz}}| \leq \frac{\varepsilon}{n+1}, \quad \|\nabla z_i\| \leq 1 + \varepsilon.$$

If $x_j \geq 1$, every path from x to E_j crosses the slab $0 \leq x_j \leq 1$. Apply (2.3) with $k = n - 1$ to that crossing.

$$\text{If } x_j < 0, \quad \varphi_j(x) \leq 0, \quad \psi_j(x) = \varphi_j(x) - \frac{\ell_j}{2} \leq -\frac{\ell_j}{2},$$

$$z_j(x) \leq -\frac{\ell_j}{2} + \frac{\varepsilon}{n+1}.$$

$$\text{If } x_j > 1, \quad \psi_j(x) \geq \frac{\ell_j}{2} - \frac{(n-1)\varepsilon}{n+1} \quad \text{by (2.3),}$$

$$z_j(x) \geq \frac{\ell_j}{2} - \frac{n\varepsilon}{n+1} > \frac{\ell_j}{2} - \varepsilon.$$

$\|\nabla z_i\| \leq 1 + \varepsilon \rightsquigarrow$ for every vector field V , $|V(z_i)| \leq (1 + \varepsilon)\|V\|$. Since $\bar{g}$ is uniformly equivalent to g_o , $\text{dist}_{\bar{g}}(x, E_i) \geq c|x_i|$ for some $c > 0$. Thus $\sum_i z_i(x)^2 \rightarrow \infty$ as $|x| \rightarrow \infty$; in particular, its sublevel sets are compact.

Step 3. Choose a positive $\delta > 0$ and $\{\bar{\ell}_i\}_{1 \leq i \leq n}$ such that $0 < \bar{\ell}_i < \ell_i - 2\varepsilon < \ell_i$,

$$\frac{n^2}{4} - \sum_{i=1}^n \frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} > \delta$$

and, for every $j = 1, \dots, n$,

$$-\frac{n\sigma}{4(n-1)} + \frac{\pi^2(1+\varepsilon)^2}{(\ell_j - 2\varepsilon)^2} \tan^2\left(\frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)}\right) - \varepsilon(1+\varepsilon) - \sum_{i \neq j} \frac{\pi^2(1+\varepsilon)^2}{(\ell_i - 2\varepsilon)^2} > \delta$$

$$\bar{\ell}_j \rightarrow \ell_j - 2\varepsilon \quad \text{from below.} \quad \text{So} \quad \frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)} \rightarrow \frac{\pi}{2} \rightsquigarrow \tan^2\left(\frac{\pi\bar{\ell}_j}{2(\ell_j - 2\varepsilon)}\right) \rightarrow \infty.$$

Set $r_i := \frac{\pi(1+\varepsilon)}{\ell_i - 2\varepsilon}$. Then $\frac{n^2}{4} - \sum_{i=1}^n r_i^2 > \delta$. Fix $\ell > 0$ such that $\ell > \ell_i/2 + 1$ for every i .

Our choice of ε gives $\varepsilon(1+\varepsilon) < r_i^2$. Choose smooth $\{\xi_i\}_{1 \leq i \leq n}$ on $\mathbb{E}^2$, even in both variables, such that

$$0 \leq (1+\varepsilon)\xi_i(x, y) \leq y^2 + r_i^2,$$

$$0 \leq \xi_i(x, y) \leq \varepsilon \quad \text{for } |x| \geq \frac{\ell_i}{2} - \varepsilon,$$

with smooth interpolation between the specified regions:

$$\xi_i(x, y) = \begin{cases} \frac{y^2 + r_i^2}{1 + \varepsilon}, & |x| \leq \frac{\bar{\ell}_i}{2}, \quad \text{near origin.} \\ 0, & \frac{\ell_i}{2} - \varepsilon \leq |x| \leq \ell, \quad \text{near exterior threshold} \\ \varepsilon, & |x| \geq \ell + 1, \quad \text{far away.} \end{cases}$$

Here $\bar{\ell}_i < \ell_i - 2\varepsilon$, so $\bar{\ell}_i/2 < \ell_i/2 - \varepsilon$. Choose the interpolation to vanish already slightly before $|x| = \ell_i/2 - \varepsilon$; this leaves a nonempty interval after $\bar{\ell}_i/2$ on which to interpolate.

$$\left\{ \begin{array}{l} f_i^2 - (1 + \varepsilon)f_i' \geq -r_i^2 \\ f_i \text{ is large near the relevant face level} \\ f_i' \text{ is small outside the cube} \\ f_i \text{ is defined on } \mathbb{R} \\ \sum_{i=1}^n f_i(z_i)^2 \text{ is proper.} \end{array} \right.$$

Let f_i be a solution of

$$\begin{cases} f'(x) = \xi_i(x, f(x)), \\ f(0) = 0, \end{cases} \quad \xi_i \geq 0 \rightsquigarrow f_i'(x) \geq 0 \Rightarrow f_i \text{ is nondecreasing.}$$

Consider $h_i(x) = r_i \tan\left(\frac{r_i x}{1 + \varepsilon}\right)$:

$$\begin{aligned} h_i'(x) &= \frac{r_i^2}{1 + \varepsilon} \sec^2\left(\frac{r_i x}{1 + \varepsilon}\right) \\ &= \frac{r_i^2}{1 + \varepsilon} \left[1 + \tan^2\left(\frac{r_i x}{1 + \varepsilon}\right) \right] \\ &= \frac{r_i^2 + h_i(x)^2}{1 + \varepsilon} \end{aligned}$$

So h_i solves

$$\begin{cases} f'(x) = \frac{f(x)^2 + r_i^2}{1 + \varepsilon} \\ f(0) = 0, \end{cases}$$

By uniqueness,

$$f_i(x) = r_i \tan\left(\frac{r_i x}{1 + \varepsilon}\right), \quad |x| \leq \frac{\bar{\ell}_i}{2}.$$

The tangent has poles at

$$|x| = \frac{\pi(1 + \varepsilon)}{2r_i} = \frac{\ell_i - 2\varepsilon}{2} = \frac{\ell_i}{2} - \varepsilon.$$

On $\ell_i/2 - \varepsilon \leq |x| \leq \ell$, $f'_i(x) = 0$. For $|x|$ large, $f'_i(x) = \varepsilon$, so f_i is linear. Comparison with h_i bounds f_i until the cutoff becomes zero, strictly before the tangent pole. It is then constant up to $|x| = \ell$, and has bounded derivative beyond that, so there is no finite-time blow-up.

$\sum_{i=1}^n f_i(z_i)^2$ is proper. The function f_i exists on $\mathbb{R}$, and for $|x|$ large it is of the form $\varepsilon x \pm c$. The evenness of ξ_i makes f_i odd. Since $f_i(x) - \varepsilon x$ is bounded and $\sum_i z_i^2$ is proper, $\sum_i f_i(z_i)^2$ is proper as well.

Step 4. $\text{Cl}_{n,0}$ is generated by $\{e_i\}_{1 \leq i \leq n}$ subject to $e_i^2 = -1$, $e_i e_j + e_j e_i = 0$, $i \neq j$.

$\text{Cl}_{0,n}$ is generated by $\{\widehat{e}_i\}_{1 \leq i \leq n}$ subject to $\widehat{e}_i^2 = 1$, $\widehat{e}_i \widehat{e}_j + \widehat{e}_j \widehat{e}_i = 0$, $i \neq j$.

Let $E := S(T\mathbb{R}^n \oplus \mathbb{E}^n)$ be equipped with the action of $\text{Cl}_{n,0}(T\mathbb{R}^n) \widehat{\otimes} \text{Cl}_{0,n}(\mathbb{E}^n)$. Here S is the graded spinor module bundle of rank 2^n , and the auxiliary bundle $\mathbb{E}^n$ has its flat connection. The connection on E is the induced tensor connection. Clifford actions from the two factors anticommute. Let D be the twisted Dirac operator on $L^2(\mathbb{R}^n, E)$. In terms of local orthonormal frames $\{w_i\}_{1 \leq i \leq n}$, $D = \sum_{i=1}^n w_i \nabla_{w_i}$.

$$B = \overset{\text{Dirac}}{D \widehat{\otimes} 1} + \sum_{i=1}^n \overset{n \text{ functions } f_i(z_i)}{1 \widehat{\otimes} f_i(z_i) \widehat{e}_i} \quad \text{Callias-type operator.}$$

$$\begin{aligned} B^2 &= (D \widehat{\otimes} 1)^2 + \sum_{i=1}^n \left[(D \widehat{\otimes} 1)(1 \widehat{\otimes} f_i(z_i) \widehat{e}_i) \right. \\ &\quad \left. + (1 \widehat{\otimes} f_i(z_i) \widehat{e}_i)(D \widehat{\otimes} 1) \right] \\ &\quad + \sum_{i,j=1}^n (1 \widehat{\otimes} f_i(z_i) \widehat{e}_i)(1 \widehat{\otimes} f_j(z_j) \widehat{e}_j) \\ &= D^2 \widehat{\otimes} 1 + \underbrace{\sum_{i=1}^n f_i(z_i)^2}_{\text{proper}} + \underbrace{\sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i}_{\text{bounded.}} \end{aligned} \quad (2.10)$$

$$\|[D, f_i(z_i)]\| = |f'_i(z_i)| \underbrace{\|\nabla z_i\|}_{\leq 1+\varepsilon}.$$

The norm here is the pointwise endomorphism norm. The commutator term is bounded. $\rightsquigarrow B$ is essentially self-adjoint and Fredholm. This uses the Callias Fredholm result. Throughout, $\text{Ind}(B)$ is the graded index: the dimension of the even kernel minus the dimension of the odd kernel.

Take $g_t = (1-t)\bar{g} + t g_o$, $0 \leq t \leq 1$; these metrics are uniformly equivalent to g_o . Use the homotopy $\theta_{t,i}(x) = (1-t)f_i(z_i(x)) + tx_i$ (potential of B). For sufficiently large $|x_i|$, both terms have the sign of x_i and grow at least linearly in $|x_i|$. Thus $\sum_i \theta_{t,i}^2$ tends to infinity uniformly in t , and all gradients are uniformly bounded. By homotopy invariance of the Fredholm index, $\text{Ind}(B) = \text{Ind}(B_o)$.

$$B_o = \sum_{i=1}^n \frac{\partial}{\partial x_i} e_i \widehat{\otimes} 1 + \sum_{i=1}^n 1 \widehat{\otimes} x_i \widehat{e}_i.$$

This operator acts on $L^2(\mathbb{R}^n, S(\mathbb{E}^n \oplus \mathbb{E}^n))$, and

$$B_o^2 = - \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2 + \underbrace{\sum_{i=1}^n e_i \widehat{\otimes} e_i}_{\substack{\{e_i \widehat{\otimes} e_i\}_{1 \leq i \leq n} \\ \text{is a commuting family of symmetries}}}.$$

$$(e_i \widehat{\otimes} e_i)^2 = 1, \\ (e_i \widehat{\otimes} e_i)(e_j \widehat{\otimes} e_j) = (e_j \widehat{\otimes} e_j)(e_i \widehat{\otimes} e_i) \quad i \neq j.$$

For $i \neq j$, these symmetries commute.

Let Λ be the set of all maps $\lambda : \{1, \dots, n\} \longrightarrow \{1, -1\}$. For each $\lambda \in \Lambda$, define V_λ by

$$\underbrace{(e_i \widehat{\otimes} e_i)v}_{\text{acts as multiplication.}} = \underbrace{\lambda(i)}_{\text{Symmetry.}} v \quad \forall i.$$

There are 2^n maps λ . $\lambda(i) \in \{-1, 1\}$. The family $\{e_i \widehat{\otimes} e_i\}_{1 \leq i \leq n}$ decomposes $S(\mathbb{E}^n \oplus \mathbb{E}^n)$ into 2^n subspaces V_λ . Clifford multiplication by e_j interchanges the eigenspaces whose j -th signs differ, leaving the other signs unchanged. Thus all ranks are equal; since $\text{rank } S(\mathbb{E}^n \oplus \mathbb{E}^n) = 2^n$, each V_λ has rank 1.

On V_λ ,

$$B_o^2 = \underbrace{- \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2}_{n\text{-dimensional harmonic oscillator}} + \sum_{i=1}^n \lambda(i).$$

In one dimension, $-\frac{d^2}{dx^2} + x^2$ has eigenvalues $O = \{1, 3, 5, \dots\}$. In n dimensions, the set of eigenvalues is $O_n := \{k_1 + k_2 + \dots + k_n : k_i \in O\}$. The smallest eigenvalue of the n -dimensional harmonic oscillator is $1 + 1 + \dots + 1 = n$. Hence the smallest eigenvalue of B_o^2 on V_λ is

$$n + \sum_{i=1}^n \lambda(i).$$

Since $\lambda(i) \in \{-1, 1\}$,

$$\sum_{i=1}^n \lambda(i) \geq -n,$$

with equality when $\lambda(i) = -1$ for every i .

If λ is not this constant map, at least one -1 is replaced by 1, and hence

$$\sum_{i=1}^n \lambda(i) \geq -n + 2.$$

Thus the smallest eigenvalue is at least $n + (-n) + 2 = 2$. For every $\lambda \neq -1$, B_o^2 has no 0 eigenvalue on $L^2(\mathbb{R}^n, V_\lambda)$.

For $\lambda \equiv -1$, $\sum_{i=1}^n \lambda(i) = -n$. Therefore,

$$B_o^2 = - \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2} + \sum_{i=1}^n x_i^2 - n.$$

Its kernel is

$$\ker B_o^2 = \left\{ \exp\left(-\frac{x_1^2 + \cdots + x_n^2}{2}\right) v : v \in \underline{V_\lambda} \right\}.$$

Since $\text{rank } V_\lambda = 1$, $\ker B_o^2$ is one-dimensional. In the standard exterior-algebra model of the graded spinor module, the common -1 eigenspace consists of scalar forms, so V_λ is even. Self-adjointness gives $\ker B_o = \ker B_o^2$. Hence

$$\text{Ind}(B_o) = \dim(\text{even kernel}) - \dim(\text{odd kernel}) = 1 - 0 = 1 = \text{Ind}(B).$$

The full operator is self-adjoint; the graded index is nonzero.

Step 5. Lichnerowicz's formula: $D^2 = \nabla^* \nabla + \frac{\text{Sc}_{\bar{g}}}{4}$. There is no auxiliary curvature term because $\mathbb{B}^n$ is flat. The operator inequalities below are quadratic-form inequalities, initially on compactly supported smooth sections; $\|v\|$ denotes the L^2 -norm.

For $v \in C_c^\infty(\mathbb{R}^n, E)$,

$$Dv = \sum_{i=1}^n w_i \nabla_{w_i} v.$$

By Cauchy–Schwarz,

$$\|Dv\|^2 \leq n \|\nabla v\|^2, \quad \|\nabla v\|^2 \geq \frac{1}{n} \|Dv\|^2.$$

$$\begin{aligned} \|Dv\|^2 &= \|\nabla v\|^2 + \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2 \\ &\geq \frac{1}{n} \|Dv\|^2 + \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2. \end{aligned}$$

$$\Rightarrow \frac{n-1}{n} \|Dv\|^2 \geq \int_{\mathbb{R}^n} \frac{\text{Sc}_{\bar{g}}}{4} |v|^2 \quad \rightsquigarrow \quad D^2 \geq \frac{\text{Sc}_{\bar{g}}}{4} \frac{n}{n-1}.$$

By (2.10),

$$\begin{aligned}
 B^2 &\geq D^2 + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n \|[D, f_i(z_i)]\| \\
 &\geq \frac{\text{Sc}_{\bar{g}}}{4} \cdot \frac{n}{n-1} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n f'_i(z_i) \|\nabla z_i\| \\
 &\geq \frac{\text{Sc}_{\bar{g}}}{4} \cdot \frac{n}{n-1} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n f'_i(z_i)(1 + \varepsilon).
 \end{aligned}$$

Divide $\mathbb{R}^n$ into two cases.

The scalar-curvature lower bound is known only on the original cube: $\text{Sc}_g \geq n(n-1)$ on I^n .

Case 1. On I^n , $\bar{g} = g$, and $\text{Sc}_{\bar{g}} = \text{Sc}_g \geq n(n-1)$, $\frac{\text{Sc}_{\bar{g}}}{4} \frac{n}{n-1} \geq \frac{n^2}{4}$.

Outside the cube, the extended metric may have negative scalar curvature.

Recall $f'_i(z_i) = \xi_i(z_i, f_i(z_i))$, $(1 + \varepsilon)f'_i(z_i) \leq f_i(z_i)^2 + r_i^2 \Rightarrow f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \geq -r_i^2$.

$$\begin{aligned}
 B^2 &\geq \frac{n^2}{4} + \sum_{i=1}^n \left(f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \right) \\
 &\geq \frac{n^2}{4} - \sum_{i=1}^n \underbrace{r_i^2}_{\frac{\pi^2(1 + \varepsilon)^2}{(\ell_i - 2\varepsilon)^2}} > \delta.
 \end{aligned}$$

Case 2. Outside the cube,

$$x \notin I^n \Rightarrow \text{for some } i, \quad |z_i(x)| \geq \frac{\ell_i}{2} - \varepsilon.$$

$$\text{Without loss of generality, } |z_1| \geq \underbrace{\frac{\ell_1}{2} - \varepsilon}_{> \bar{\ell}_1/2}.$$

Outside: positivity comes from the potential.

On $|x| \leq \bar{\ell}_1/2$, $f_1(x) = r_1 \tan\left(\frac{r_1 x}{1 + \varepsilon}\right)$. Since f_1 is nondecreasing, for $|x| \geq \ell_1/2 - \varepsilon$ we have

$$\begin{aligned}
 |f_1(x)| &\geq \underbrace{r_1}_{\frac{\pi(1 + \varepsilon)}{\ell_1 - 2\varepsilon}} \tan\left(\frac{r_1 \bar{\ell}_1}{2(1 + \varepsilon)}\right)
 \end{aligned}$$

Also, $|f'_1(z_1)| \leq \varepsilon$. $\text{Sc}_{\bar{g}} \geq -\sigma$. For $i \geq 2$, $f_i(z_i)^2 - (1 + \varepsilon)f'_i(z_i) \geq -r_i^2$.

$$\begin{aligned} B^2 &\geq -\frac{n\sigma}{4(n-1)} + f_1(z_1)^2 - \varepsilon(1 + \varepsilon) - \sum_{i=2}^n r_i^2 \\ &\geq -\frac{n\sigma}{4(n-1)} + \frac{\pi^2(1 + \varepsilon)^2}{(\ell_1 - 2\varepsilon)^2} \tan^2\left(\frac{\pi\bar{\ell}_1}{2(\ell_1 - 2\varepsilon)}\right) - \varepsilon(1 + \varepsilon) - \sum_{i=2}^n r_i^2 \\ &> \delta. \end{aligned}$$

Thus $\|Bv\|^2 \geq \delta\|v\|^2$ on the domain of B , by closure from the smooth core. Since B is self-adjoint, B is invertible. $(\text{Ran } B)^\perp = \ker B = 0$, so the range is dense and closed. An invertible graded Fredholm operator has no kernel, so $\text{Ind}(B) = 0$. ∇

Part II: strict inequality

To prove strictness under the dihedral angle assumption, we need to exclude

$$\underbrace{\sum_{i=1}^n \frac{1}{\ell_i^2}}_{\text{assume to the contrary.}} = \frac{n^2}{4\pi^2}.$$

Step 1. Let $E := S(TI^n \oplus \mathbb{E}^n)$ be the bundle over I^n , and set

$$B := D \widehat{\otimes} 1 - \sum_{i=1}^n 1 \widehat{\otimes} \underbrace{f_i(z_i)}_{\substack{\text{to be specified} \\ \text{later}}} \widehat{e}_i.$$

Here D and the flat auxiliary connection are as in Part I, restricted to I^n . The minus sign in this part matches the inward-normal boundary condition and the negative tangent functions below. For now, assume the compositions $f_i(z_i)$ are smooth on the closed cube.

Boundary: the tangent term is very large near the faces; the boundary argument ultimately forces a kernel section to vanish on ∂I^n .

Choose a manifold with corners $M \subset \mathbb{R}^n$ and an orientation-preserving corner diffeomorphism $\Phi : I^n \rightarrow M$ as follows. For sufficiently large $R > 1$, set

$$M = \bigcap_{i=1}^n (\overline{B}(\widehat{e}_i, R) \cap \overline{B}(-\widehat{e}_i, R)), \quad \Phi(x) = \frac{(R^2 - 1)(2x_1 - 1, \dots, 2x_n - 1)}{\sqrt{1 + (R^2 - 1) \sum_{j=1}^n (2x_j - 1)^2 + 1}}.$$

Here $\overline{B}(y, R)$ is the closed Euclidean ball of radius R centered at y . The inverse has coordinates $x_i = \frac{1}{2} + y_i/(R^2 - 1 - |y|^2)$, so this is a smooth diffeomorphism. The faces $\Phi(\partial_{i\pm})$ lie on the spheres centered at $\mp \widehat{e}_i$.

(1) Φ preserves the corner structures.

$$\begin{array}{ccc} \text{codimensions } 1, 2 & & \text{codimensions } 1, 2 \\ \downarrow & & \downarrow \\ \text{face of } I^n & \longrightarrow & \text{face of } M \end{array}$$

(2) At each codimension-2 face,

$$0 < \underbrace{\theta(x)}_{\text{dihedral angle of } (I^n, g)} < \underbrace{\theta_M(\Phi(x))}_{\text{of } M} < \pi$$

Indeed, every dihedral angle of M is $\pi - \arccos(1 - R^{-2})$, which tends to π as R tends to infinity. The angles of (I^n, g) have maximum strictly less than π , so such an R exists.

↪ Use index theory for manifolds with corners.

(3) On any closed codimension-one face F of I^n , use the inward normal to $\Phi(F)$. Its components satisfy

$$\begin{aligned} \langle \widehat{\nu}, \widehat{e}_i \rangle &< 0 \quad \text{on } \Phi(F \cap \partial_{i+}), \\ \langle \widehat{\nu}, \widehat{e}_i \rangle &> 0 \quad \text{on } \Phi(F \cap \partial_{i-}). \end{aligned}$$

At $y \in \Phi(\partial_{j\pm})$, this normal is $-(y \pm \widehat{e}_j)/R$. The formula for Φ shows the asserted signs for every incident coordinate, including at corners.

On each codimension-one face $F \subset I^n$, the boundary condition $\mathfrak{B}$ is given by

$$(\nu_x \widehat{\otimes} \widehat{\nu}_{\Phi(x)})\varphi(x) = -\varphi(x), \quad \Phi(F) \subset M.$$

Here ν_x is the inward unit normal to F . The corner index theorem applies: both tangent bundles are trivial, so Φ is spin. It has degree 1, and M is diffeomorphic to the cube, so its Euler characteristic is 1. The smooth potential is a bounded self-adjoint zeroth-order perturbation, so the self-adjoint domain is unchanged. It is compact as a map $H^1 \rightarrow L^2$, so the Fredholm index is unchanged. Hence:

- (1) B with $\mathfrak{B}$ is self-adjoint and Fredholm, with domain $H^1(I^n, E; \mathfrak{B})$.
- (2) $\text{Ind}(B) = 1$. Therefore, there exists a **nonzero** $\varphi \in H^1(I^n, E; \mathfrak{B})$ such that $B\varphi = 0$.

Step 2. Bochner formula. Let ψ be a smooth section of E over I^n that vanishes near all codimension-2 faces and satisfies $\mathfrak{B}$ on every codimension-one face.

$$\begin{aligned} \int_{I^n} \langle B\psi, B\psi \rangle &= \int_{I^n} |D\psi|^2 \\ &+ \int_{I^n} \left\langle \left(\sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n \underbrace{[D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i}_{\text{commutator}} \right) \psi, \psi \right\rangle \\ &- \underbrace{\int_{\partial I^n}}_{\text{boundary}} \sum_{i=1}^n \langle f_i(z_i) (\nu \widehat{\otimes} \widehat{e}_i) \psi, \psi \rangle. \end{aligned}$$

Part I: $|D\psi|^2 \leq n|\nabla\psi|^2 \longrightarrow$ discards information in the equality case.

Penrose operator. $P_V\psi := \nabla_V\psi + \frac{1}{n}V \cdot D\psi, \quad V \in TI^n.$

$$|\nabla\psi|^2 = \underbrace{|P\psi|^2}_{\geq 0} + \underbrace{\frac{1}{n}|D\psi|^2}_{\geq 0}. \quad \rightsquigarrow \quad |D\psi|^2 \leq n|\nabla\psi|^2$$

↘ Records the equality condition in the Lichnerowicz formula.

The Lichnerowicz formula gives

$$\begin{aligned} \int_{I^n} |D\psi|^2 &= \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |\nabla\psi|^2 \right) + \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle \\ \underbrace{\left(1 - \frac{1}{n} \right)}_{\frac{n-1}{n}} \int_{I^n} |D\psi|^2 &= \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |P\psi|^2 \right) + \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle. \\ \Rightarrow \int_{I^n} |D\psi|^2 &= \frac{n}{n-1} \int_{I^n} \left(\frac{\text{Sc}_g}{4} |\psi|^2 + |P\psi|^2 \right) + \frac{n}{n-1} \int_{\partial I^n} \langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle \end{aligned}$$

On each codimension-one face of I^n ,

$$\nu \cdot D + \nabla_\nu = \sum_{j=1}^{n-1} \nu \cdot e_j \cdot \nabla_{e_j},$$

where $\{e_j\}_{j=1}^{n-1}$ is a tangent orthonormal frame.

Define

$$\begin{aligned} \mathcal{H} := -\frac{1}{2} &\left[(\nu \cdot D + \nabla_\nu)(\nu \widehat{\otimes} \widehat{\nu}) \right. \\ &\left. + (\nu \widehat{\otimes} \widehat{\nu})(\nu \cdot D + \nabla_\nu) \right]. \end{aligned}$$

The first-order terms cancel because the tangential Clifford symbols anticommute with $\nu \widehat{\otimes} \widehat{\nu}$. Thus $\mathcal{H}$ is a bounded endomorphism. The boundary condition gives

$$\langle (\nu \cdot D + \nabla_\nu)\psi, \psi \rangle = \langle \mathcal{H}\psi, \psi \rangle.$$

Integration by parts on a manifold with boundary shows that $\nu \cdot D + \nabla_\nu$ is not automatically nonnegative.

At a point of a face of M , decompose

$$\widehat{e}_i = \underbrace{a_i}_{\langle \widehat{\nu}, \widehat{e}_i \rangle} \widehat{\nu} + \underbrace{\widehat{w}_i}_{\perp \widehat{\nu}}.$$

$$\langle (\nu \widehat{\otimes} \widehat{e}_i) \psi, \psi \rangle = a_i \langle (\nu \widehat{\otimes} \widehat{\nu}) \psi, \psi \rangle + \underbrace{\langle (\nu \widehat{\otimes} \widehat{w}_i) \psi, \psi \rangle}_{\substack{\text{anticommutes} \\ \swarrow = 0}}$$

The boundary condition makes the first term $-a_i |\psi|^2$. The second vanishes because its operator interchanges the two eigenspaces of the self-adjoint involution $\nu \widehat{\otimes} \widehat{\nu}$.

The following identity holds:

$$\begin{aligned} \int_{I^n} \langle B\psi, B\psi \rangle &= \frac{n}{n-1} \int_{I^n} |P\psi|^2 \\ &+ \int_{I^n} \left\langle \left(\frac{n}{n-1} \frac{\text{Sc}_g}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i \right) \psi, \psi \right\rangle \\ &+ \frac{n}{n-1} \int_{\partial I^n} \langle \mathcal{H}\psi, \psi \rangle \\ &+ \int_{\partial I^n} \left(\sum_{i=1}^n \langle \widehat{\nu}, \widehat{e}_i \rangle f_i(z_i) \right) |\psi|^2 \end{aligned} \quad (\star)$$

This holds for every $\psi \in H^1(I^n, E; \mathfrak{B})$. Indeed, smooth sections satisfying the boundary condition and vanishing near the codimension-two faces are dense in this H^1 -space. Both sides are H^1 -continuous, including the boundary terms by the trace theorem.

Step 3. Define $f_i(x) = -\frac{\pi}{\ell_i} \tan\left(\frac{\pi x}{\ell_i}\right)$ $x \in (-\ell_i/2, \ell_i/2)$.

$$\begin{aligned} |f'_i(x)| &= \left| -\frac{\pi}{\ell_i} \frac{\pi}{\ell_i} \sec^2\left(\frac{\pi x}{\ell_i}\right) \right| \\ &= \frac{\pi^2}{\ell_i^2} \underbrace{\sec^2\left(\frac{\pi x}{\ell_i}\right)}_{1 + \tan^2\left(\frac{\pi x}{\ell_i}\right)} \\ f_i^2(x) &= \frac{\pi^2}{\ell_i^2} \tan^2\left(\frac{\pi x}{\ell_i}\right). \end{aligned}$$

$$\rightsquigarrow |f'_i(x)| = \frac{\pi^2}{\ell_i^2} + f_i^2(x) \quad \Rightarrow \quad \frac{\pi^2}{\ell_i^2} + f_i^2 - |f'_i| = 0, \quad f_i(x) \longrightarrow \mp\infty, \quad x \longrightarrow \pm\ell_i/2.$$

Recall that $\ell_i = \text{dist}_g(\partial_{i-}, \partial_{i+})$. Start with the 1-Lipschitz function

$$x \longmapsto \min\{\ell_i, \text{dist}_g(x, \partial_{i-})\} - \frac{\ell_i}{2}.$$

$$\begin{aligned} \text{on } \partial_{i-}, \quad & \rightsquigarrow 0 \quad \Rightarrow \quad -\frac{\ell_i}{2} \\ \text{on } \partial_{i+}, \quad & \rightsquigarrow \ell_i \quad \Rightarrow \quad \frac{\ell_i}{2}. \end{aligned}$$

For a choice that is strictly between these endpoint values away from the two faces, set

$$\widetilde{z}_i(x) = \frac{\ell_i \operatorname{dist}_g(x, \partial_{i-}) - \operatorname{dist}_g(x, \partial_{i+})}{2 \operatorname{dist}_g(x, \partial_{i-}) + \operatorname{dist}_g(x, \partial_{i+})}.$$

The denominator is at least ℓ_i , by the triangle inequality. Since both distance functions are 1-Lipschitz, differentiation almost everywhere gives

$$\|\nabla \widetilde{z}_i(x)\| \leq \frac{\ell_i}{\operatorname{dist}_g(x, \partial_{i-}) + \operatorname{dist}_g(x, \partial_{i+})} \leq 1.$$

Thus $\widetilde{z}_i$ is 1-Lipschitz, equals $\pm \ell_i/2$ on $\partial_{i\pm}$, and satisfies $|\widetilde{z}_i| < \ell_i/2$ elsewhere.

On each closed face, write $a_i = \langle \widehat{\nu}, \widehat{e}_i \rangle$, using that face's normal. For $0 < t < 1$, consider $(1-t)\widetilde{z}_i$. On $\Phi(\partial_{i-})$, $a_i > 0$, and for $x \in \partial_{i-}$,

$$\begin{aligned} (1-t)\widetilde{z}_i(x) &= -(1-t)\frac{\ell_i}{2} \longrightarrow -\frac{\ell_i}{2}, \quad f_i((1-t)\widetilde{z}_i(x)) \longrightarrow +\infty, \\ \Rightarrow \quad a_i f_i((1-t)\widetilde{z}_i(x)) &\longrightarrow +\infty \quad (t \downarrow 0, x \in \partial_{i-}). \end{aligned}$$

On $\Phi(\partial_{i+})$, $a_i < 0$, and for $x \in \partial_{i+}$,

$$\begin{aligned} (1-t)\widetilde{z}_i(x) &= (1-t)\frac{\ell_i}{2} \longrightarrow \frac{\ell_i}{2}, \quad f_i((1-t)\widetilde{z}_i(x)) \longrightarrow -\infty, \\ \Rightarrow \quad a_i f_i((1-t)\widetilde{z}_i(x)) &\longrightarrow +\infty \quad (t \downarrow 0, x \in \partial_{i+}). \end{aligned}$$

Claim 2.3. $\sum_{i=1}^n a_i(x) f_i((1-t)\widetilde{z}_i(x)) \longrightarrow +\infty \quad (t \downarrow 0)$ uniformly on each closed codimension-one face.

Fix $F = \partial_{j\pm}$. Its j -th term tends uniformly to $+\infty$, since $\widetilde{z}_j = \pm \ell_j/2$ there and $|a_j|$ has a positive minimum. For $i \neq j$, near $F \cap \partial_{i\pm}$, the incident-face sign condition makes $a_i f_i((1-t)\widetilde{z}_i)$ nonnegative for sufficiently small t . Away from these intersections, $\widetilde{z}_i$ stays uniformly away from $\pm \ell_i/2$, so this term is uniformly bounded. Thus the remaining terms are uniformly bounded below, and the j -th term dominates them. Compactness of F , and the finite number of faces, give the claim.

Step 4. First fix a sufficiently small $t > 0$ and replace $\widetilde{z}_i$ by $(1-t)\widetilde{z}_i$. The claim gives a strict boundary margin. This contraction also leaves a Lipschitz constant at most $1-t$ and keeps the range away from the poles. Smooth sufficiently closely, with an increase in Lipschitz constant smaller than t , to obtain smooth 1-Lipschitz functions z_i . At corners, use local inward translations before convolution and then a partition of unity; the increase in Lipschitz constant can still be made arbitrarily small. The strict range and boundary

inequalities persist: $|z_i(x)| < \ell_i/2$ for every $x \in I^n$, and

$$\begin{aligned} &\text{For every } x \in \partial_{j\pm}, \quad 1 \leq j \leq n, \\ &\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i(x)) > \frac{n}{n-1} \|\mathcal{H}\|. \end{aligned} \quad \leftarrow \text{perturbation near the boundary}$$

Since $\text{Sc}_g \geq n(n-1)$,

$$\frac{n}{n-1} \frac{\text{Sc}_g}{4} \geq \frac{n^2}{4}.$$

By 1-Lipschitzness, $\|\nabla z_i\| \leq 1$, so

$$\begin{aligned} \|[D, f_i(z_i)]\| &= |f'_i(z_i)| \|\nabla z_i\| \leq |f'_i(z_i)|, \\ -[D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i &\geq -\|[D, f_i(z_i)]\| \geq -|f'_i(z_i)|. \end{aligned}$$

Using $f_i(z_i)^2 - |f'_i(z_i)| = -\pi^2/\ell_i^2$, we obtain

$$\begin{aligned} &\frac{n}{n-1} \frac{\text{Sc}_g}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n [D, f_i(z_i)] \widehat{\otimes} \widehat{e}_i \\ &\geq \frac{n^2}{4} + \sum_{i=1}^n f_i(z_i)^2 - \sum_{i=1}^n |f'_i(z_i)| \\ &= \frac{n^2}{4} - \underbrace{\sum_{i=1}^n \frac{\pi^2}{\ell_i^2}}_{=\frac{n^2}{4}} = 0. \end{aligned}$$

Let $\varphi \in H^1(I^n, E; \mathfrak{B})$ be the nonzero section from Step 1 such that $B\varphi = 0$. By $(\star)$, $\int_{I^n} \langle B\varphi, B\varphi \rangle = 0$.

$$\begin{aligned} \frac{n}{n-1} \langle \mathcal{H}\varphi, \varphi \rangle + \left(\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i) \right) |\varphi|^2 &\geq \left(\sum_{i=1}^n \langle \widehat{v}, \widehat{e}_i \rangle f_i(z_i) - \frac{n}{n-1} \|\mathcal{H}\| \right) |\varphi|^2 \\ &\geq 0. \end{aligned}$$

The coefficient in parentheses is strictly positive on ∂I^n . Since the interior terms and the combined boundary term in $(\star)$ are nonnegative, equality gives

$$\int_{I^n} |P\varphi|^2 = 0 \quad \Rightarrow \quad P\varphi = 0 \quad \text{almost everywhere on } I^n.$$

$\varphi = 0$ almost everywhere on ∂I^n .

Since $B\varphi = 0$ and $P\varphi = 0$,

$$\begin{aligned} \nabla_V \varphi + \frac{1}{n} V \cdot D\varphi &= 0 \quad \Rightarrow \quad \nabla_V \varphi = -\frac{1}{n} V \cdot D\varphi, \\ D\varphi - \sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi &= 0 \quad \Rightarrow \quad D\varphi = \sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi, \\ \nabla_V \varphi &= -\frac{1}{n} V \cdot \left(\sum_{i=1}^n f_i(z_i) \widehat{e}_i \varphi \right) \\ &= -\frac{1}{n} \left(\sum_{i=1}^n V \widehat{\otimes} f_i(z_i) \widehat{e}_i \right) \varphi, \quad \text{for every } V \in TI^n. \end{aligned}$$

In a smooth local trivialization, this expresses each weak first derivative of φ as a smooth matrix times φ . Iteration gives H^k -regularity for every k , hence smoothness up to the boundary by Sobolev embedding. Its zero trace therefore gives $\varphi(y_0) = 0$ for any $y_0 \in \partial I^n$.

Let $x \in I^n$. Choose a smooth path γ_x connecting x to y_0 . Let $V = \dot{\gamma}_x$.

$$\nabla_{\dot{\gamma}_x} \varphi = -\frac{1}{n} \left(\sum_{i=1}^n \dot{\gamma}_x \widehat{\otimes} f_i(z_i) \widehat{e}_i \right) \varphi.$$

By uniqueness, $\varphi(y_0) = 0$ implies that $\varphi = 0$ along γ_x . Hence $\varphi(x) = 0$. Since $x \in I^n$ was arbitrary, $\varphi = 0$ on I^n , a contradiction. $\square$

3 Gromov's simplicial volume vanishing conjecture for positive scalar curvature and fundamental group decay

Gromov's conjecture ([MY26, (1.2)]). *Let M be closed and oriented. If M admits positive scalar curvature (PSC), then $\| [M] \|_{\ell^1} = 0$.*

Assumptions. The universal cover $\tilde{M}$ is spin, and the fundamental group has sufficiently small decay rate.

3.1 Scalar curvature and simplicial volume

3.1.1 Scalar curvature

At a point $p \in M$, scalar curvature is characterized by

$$\frac{\text{Vol}(B_M(p, r))}{\text{Vol}(B_{\mathbb{R}^m}(r))} = 1 - \frac{\text{Scal}(p)}{6(m+2)} r^2 + o(r^2), \quad \frac{o(r^2)}{r^2} \rightarrow 0 \quad (r \rightarrow 0).$$

PSC: very small balls have smaller volume than Euclidean balls. Moreover,

$$\text{Scal}(p) = 6(m+2) \lim_{r \rightarrow 0} \frac{1 - \text{Vol}(B_M(p, r)) / \text{Vol}(B_{\mathbb{R}^m}(0, r))}{r^2}.$$

Thus, if $\text{Scal}(p) > 0$, then $\text{Vol}(B_M(p, r)) < \text{Vol}(B_{\mathbb{R}^m}(0, r))$ for sufficiently small r .

The normalization is $\text{Scal}(S^m) = m(m-1) \rightsquigarrow \text{Scal} \geq m(m-1)$.

Theorem 3.1 (Lichnerowicz [MY26, (1.1)]). *If M is closed and spin and M admits PSC, then $\widehat{A}(M) = 0$.*

3.1.2 Simplicial volume

Let M be a closed oriented m -manifold. A singular chain $c = \sum_i a_i \sigma_i$ represents the fundamental class $[M] \in H_m(M; \mathbb{R})$ if $[c] = [M]$, where $a_i \in \mathbb{R}$ and $\sigma_i: \Delta^m \rightarrow M$ is a continuous singular simplex. One may triangulate the manifold. Here

$$\Delta^m = \left\{ (t_0, \dots, t_m) \in \mathbb{R}^{m+1} : t_j \geq 0, \sum_{j=0}^m t_j = 1 \right\}.$$

Definition 3.2 (Simplicial volume [MY26, (1.3)]). The simplicial volume of M is

$$\begin{aligned} \|M\| &= \|[M]\|_{\ell^1} = \inf_{[c]=[M]} \sum_i |a_i| \\ &= \inf \left\{ \sum_i |a_i| : c = \sum_i a_i \sigma_i, \partial c = 0, [c] = [M] \right\}, \end{aligned}$$

where $\|c\|_{\ell^1} = \sum_i |a_i|$. It measures how efficiently the fundamental class can be built from singular simplices.

Intuition. Let $d \gg 1$ be a positive integer. Define $\sigma_d: [0, 1] \rightarrow S^1$ by $\sigma_d(t) = e^{2\pi i d t}$. Then $\sigma_d(0) = 1 = \sigma_d(1)$ and $\partial \sigma_d = \sigma_d(1) - \sigma_d(0) = 0$. Thus σ_d is a singular 1-cycle which winds d times

around a circle. Since $[\sigma_d] = d[S^1]$, we have $\frac{1}{d}[\sigma_d] = [S^1]$ and $\|\frac{1}{d}\sigma_d\|_{\ell^1} = \frac{1}{d}$. Therefore $\|S^1\| \leq \frac{1}{d}$ for every positive integer d , so $\|S^1\| = 0$, but $[S^1] \neq 0$.

More generally, for S^n , $n \geq 1$, use self-maps $f_d: S^n \rightarrow S^n$ of degree $d \geq 1$. Then $\|f_{d*}[S^n]\|_{\ell^1} \leq \|[S^n]\|_{\ell^1}$. Since $f_{d*}[S^n] = d[S^n]$, it follows that $d\|S^n\| \leq \|S^n\|$, and hence $\|S^n\| = 0$.

Example 3.3 (Hyperbolic surfaces). Let Σ_g be a closed oriented surface of genus $g \geq 2$. Equip Σ_g with a hyperbolic metric of curvature -1 , so $K = -1$. Let ω be the area form. Then $\text{Area}(\Sigma_g) = \int_{\Sigma_g} \omega$ and $K\omega = -\omega$. By Gauss–Bonnet, $\int_{\Sigma_g} K\omega = 2\pi\chi(\Sigma_g)$. Hence

$$\int_{\Sigma_g} K\omega = - \int_{\Sigma_g} \omega = -\text{Area}(\Sigma_g) = 2\pi\chi(\Sigma_g),$$

and therefore $\text{Area}(\Sigma_g) = -2\pi\chi(\Sigma_g)$.

Let $c = \sum_i a_i \sigma_i$ be a singular fundamental cycle, and let $p: \mathbb{H}^2 \rightarrow \Sigma_g$ be the universal covering map. The metric on Σ_g lifts to a hyperbolic metric on $\mathbb{H}^2$.

For $\sigma_i: \Delta^2 \rightarrow \Sigma_g$, choose a lift $\tilde{\sigma}_i: \Delta^2 \rightarrow \mathbb{H}^2$ such that $p \circ \tilde{\sigma}_i = \sigma_i$.

Idea: replace $\tilde{\sigma}_i$ by the geodesic triangle with the same vertices and project back to Σ_g . This gives the straightening of σ_i . Set $\text{st}(c) = \sum_i a_i \text{st}(\sigma_i)$. It represents the same fundamental class: $[\text{st}(c)] = [c] = [\Sigma_g]$. Thus

$$\begin{aligned} \text{Area}(\Sigma_g) &= \langle \omega, \text{st}(c) \rangle \\ &= \left| \sum_i a_i \int_{\text{st}(\sigma_i)} \omega \right| \\ &\leq \sum_i |a_i| \left| \int_{\text{st}(\sigma_i)} \omega \right| \\ &\leq \pi \sum_i |a_i|. \end{aligned}$$

Therefore $\sum_i |a_i| \geq \text{Area}(\Sigma_g)/\pi$. Taking the infimum, $\|\Sigma_g\| \geq \text{Area}(\Sigma_g)/\pi = -2\chi(\Sigma_g) > 0$.

(Gromov) If M is a closed hyperbolic manifold, then $\|M\| = \text{Vol}(M)/v_m > 0$, where v_m is the volume of a regular ideal simplex in $\mathbb{H}^m$. For $m \geq 3$, a homotopy equivalence between closed hyperbolic m -manifolds is homotopic to an isometry.

(Gromov) If M has amenable fundamental group, then $\|M\| = 0$.

The straightening method gives the non-vanishing of simplicial volume (Thurston).

(Gromov) PSC conjecture on simplicial volume: PSC forces the simplicial volume $\|M\|$ to vanish.

Recall: S^n admits PSC and $\|S^n\| = 0$. A closed hyperbolic surface satisfies $\|\Sigma_g\| > 0$, and its hyperbolic metric has negative scalar curvature.

3.2 The Ma–Yu theorem and fundamental group decay

Theorem 3.4 (Ma–Yu [MY26, Theorem 1.1 and the following discussion]). *There exists a universal constant $C > 0$, dependent only on the dimension m , such that if M is an m -dimensional closed oriented Riemannian manifold whose universal cover $\tilde{M}$ is spin and which satisfies $\text{Scal} \geq C$ and $\text{Ric} \geq -1$, then $\|M\| = 0$.*

The lower bound $\text{Ric} \geq -1$ means that $\text{Ric}_p(v, v) \geq -g_p(v, v) = -|v|^2$ for $p \in M$ and $v \in T_p M$. If v is a unit vector and $(v, e_2, \dots, e_m)$ is an orthonormal basis of $T_p M$, then $\text{Ric}_p(v, v) = K_p(v, e_2) + \dots + K_p(v, e_m) \geq -1$.

Definition 3.5 (Volume entropy [MY26, (1.6)]). The volume entropy is $h_g = \limsup_{R \rightarrow \infty} \frac{\ln \text{Vol}(B_{\widetilde{M}}(x, R))}{R}$.

If $\text{Vol}(B(x, R)) \sim R^m$, then $h = 0$. If $\text{Vol}(B(x, R)) \sim e^{aR}$, then $h = a$.

The Ricci curvature lower bound may be replaced by an upper bound 1 on h_g .

3.2.1 Fundamental group decay

Let $x_b \in \widetilde{M}$ and $\Gamma = \pi_1(M)$. Define $d_\Gamma(\gamma, \gamma') = d_{\widetilde{M}}(\gamma x_b, \gamma' x_b)$, which is inherited from the Riemannian metric. Equivalently, the corresponding length function is $\ell(\gamma) = d_{\widetilde{M}}(x_b, \gamma x_b)$, and $B_\Gamma(R) = B(1_\Gamma, R) = \{\gamma \in \Gamma : \ell(\gamma) \leq R\}$.

For $a \in C_c(\Gamma)$, the function $a : \Gamma \rightarrow \mathbb{C}$ has finite support, $a = \sum_{\gamma \in \Gamma} a(\gamma)\gamma$, and $\|a\|_{\ell^2(\Gamma)}^2 = \sum_{\gamma \in \Gamma} |a(\gamma)|^2$. If $(a * b)(\gamma) = \sum_{\gamma' \gamma'' = \gamma} a(\gamma')b(\gamma'')$, then $\|a\|_{C_r^*(\Gamma)} = \sup_{0 \neq b \in C_c(\Gamma)} \frac{\|a * b\|_2}{\|b\|_2}$.

Definition 3.6 (Exponential decay property). The fundamental group Γ is said to satisfy the exponential decay property at rate $d \geq 0$ if there exists a constant $C > 0$ such that, for every $R > 0$ and every $a \in C_c(\Gamma)$,

$$\|a\|_{C_r^*(\Gamma)}^2 \leq C e^{2dR} \sum_{\gamma \in \Gamma} |a(\gamma)|^2$$

whenever $\text{supp}(a) \subseteq B_\Gamma(R)$. Equivalently, $\|a\|_{C_r^*(\Gamma)} \leq \sqrt{C} e^{dR} \|a\|_2$.

Remark 3.7. Replace the exponential function by a polynomial function. This gives the classical property RD.

Define a convolution operator on $\ell^2(\Gamma)$ by $(\lambda(a)\xi)(x) = \sum_{\gamma \in \Gamma} a(\gamma)\xi(\gamma^{-1}x)$. Then $\|a\|_{C_r^*(\Gamma)} = \|\lambda(a)\|_{\mathcal{B}(\ell^2(\Gamma))}$. The ratio $\|a\|_{C_r^*(\Gamma)} / \|a\|_{\ell^2(\Gamma)}$ measures how large convolution by a can be compared with the size of its coefficients.

Definition 3.8 (Decay function and decay rate [MY26, (1.12)]). Define the decay function $\Psi_\Gamma(R)$ and the decay rate ψ_Γ by

$$\Psi_\Gamma(R) = \log \left(\sup_{\substack{0 \neq a \in C_c(\Gamma) \\ \text{supp}(a) \subseteq B(1_\Gamma, R)}} \frac{\|a\|_{C_r^*(\Gamma)}}{\|a\|_{\ell^2(\Gamma)}} \right), \quad \psi_\Gamma = \limsup_{R \rightarrow \infty} \frac{\Psi_\Gamma(R)}{R}.$$

Every $d > \psi_\Gamma$ is an admissible exponential decay rate.

Definition 3.9 (Property RD [MY26, (1.20)]). Property RD means that there exist constants $C > 0$ and $s > 0$ such that, for every $R > 0$ and every $a \in C_c(\Gamma)$ with $\text{supp}(a) \subseteq B(1_\Gamma, R)$, one has $\|a\|_{C_r^*(\Gamma)} \leq C(1 + R)^s \|a\|_{\ell^2(\Gamma)}$.

Therefore

$$\sup_{\substack{0 \neq a \in C_c(\Gamma) \\ \text{supp}(a) \subseteq B(1_\Gamma, R)}} \frac{\|a\|_{C_r^*(\Gamma)}}{\|a\|_{\ell^2(\Gamma)}} \leq C(1 + R)^s,$$

and

$$\frac{\Psi_\Gamma(R)}{R} \leq \frac{\log C}{R} + s \frac{\log(1+R)}{R}.$$

Since $\frac{\log C}{R} \rightarrow 0$ and $\frac{\log(1+R)}{R} \rightarrow 0$, we obtain $\psi_\Gamma \leq 0$.

If $a = \delta_{1_\Gamma}$, then $\|a\|_{C_r^*(\Gamma)} = \|a\|_{\ell^2(\Gamma)} = 1$. Thus $\Psi_\Gamma(R) \geq 0$, so $\psi_\Gamma \geq 0$. Consequently, $\psi_\Gamma = 0$.

By Cauchy–Schwarz, $\|a\|_{C_r^*(\Gamma)} \leq \|a\|_1 \leq |B_\Gamma(R)|^{1/2} \|a\|_2$. Hence $\psi_\Gamma \leq \limsup_{R \rightarrow \infty} \frac{\log |B_\Gamma(R)|}{2R}$. Because M is compact, orbit growth can be compared with volume growth: $|B_\Gamma(R)| \leq C \text{Vol}(B_{\tilde{M}}(x, R+C))$. Therefore $\psi_\Gamma \leq h_g/2$.

Theorem 3.10 (Main theorem [MY26, Theorem 1.2 and the preceding discussion]). *There exists a universal constant $C > 0$, depending only on the dimension m , such that if M is a closed oriented manifold of dimension m whose universal cover $\tilde{M}$ is spin, and M admits a Riemannian metric with scalar curvature $\text{Scal} \geq C$ and decay rate $\psi_\Gamma \leq 1$, then $\|M\| = 0$.*

The spinor bundle is $S_{\tilde{M}}$, and the Dirac operator is on $\tilde{M}$.

3.3 Dirac operators

Let $D = C_1 \frac{\partial}{\partial x_1} + \cdots + C_m \frac{\partial}{\partial x_m}$, where $C_i^2 = -1$ and $C_i C_j + C_j C_i = 0$ for $i \neq j$. Then $D^2 = -\sum_{i=1}^m \frac{\partial^2}{\partial x_i^2} = \Delta$, the Laplacian. This is a Dirac operator retaining the Clifford algebra structure.

Let $e_1, \dots, e_m$ be an orthonormal frame. The spin Dirac operator is $D = \sum_{j=1}^m c(e_j) \nabla_{e_j}$, where $c(e_j)$ denotes Clifford multiplication. It differentiates spinors in the directions e_j .

The Lichnerowicz formula is $D^2 = \nabla^* \nabla + \text{Scal}/4$, where $\nabla^* \nabla \geq 0$. If $\text{Scal} \geq k > 0$, then for every smooth compactly supported spinor s ,

$$\langle D^2 s, s \rangle = \|\nabla s\|^2 + \frac{1}{4} \int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x). \quad (*)$$

Here $\|s\|^2 = \int_{\tilde{M}} |s(x)|^2 dv(x)$ and $\|\nabla s\|^2 = \int_{\tilde{M}} |\nabla s(x)|^2 dv(x)$, where $|\nabla s(x)|^2 = \sum_{j=1}^m |\nabla_{e_j} s(x)|^2$. Indeed, $\langle D^2 s, s \rangle = \langle \nabla^* \nabla s, s \rangle + \frac{1}{4} \langle \text{Scal} s, s \rangle$, where $\langle \text{Scal} s, s \rangle = \int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x)$. Since

$$\int_{\tilde{M}} \text{Scal}(x) |s(x)|^2 dv(x) \geq k \int_{\tilde{M}} |s(x)|^2 dv(x) = k \|s\|^2,$$

the identity $(*)$ gives

$$\langle D^2 s, s \rangle \geq 0 + \frac{k}{4} \|s\|^2 = \frac{k}{4} \|s\|^2.$$

Thus

$$D^2 \geq \frac{k}{4}, \quad \text{spec}(D) \cap \left(-\frac{\sqrt{k}}{2}, \frac{\sqrt{k}}{2} \right) = \emptyset.$$

Therefore D is invertible.

For the chiral spin Dirac operator D^+ on a closed even-dimensional spin manifold, write

$$S = S^+ \oplus S^-, \quad D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix},$$

where $D^+ : C^\infty(M, S^+) \longrightarrow C^\infty(M, S^-)$. Then $\text{Index}(D^+) = \dim \ker D^+ - \dim \ker D^-$.

By the Atiyah–Singer index theorem,

$$\text{Index}(D^+) = \widehat{A}(M) = \left\langle \widehat{A}^{[m]}(TM), [M] \right\rangle, \quad [M] \in H_m(M).$$

PSC makes D invertible, so $\ker D^+ = \ker D^- = 0$. Thus $\widehat{A}(M) = \text{Index}(D^+) = 0$.

3.4 Quantitative higher index

Let $\widetilde{D}$ be the Dirac operator on the universal cover $\widetilde{M}$. Choose smooth functions $\varphi_{\varepsilon,0}$ and $\varphi_{\varepsilon,1}$ such that $\varphi_{\varepsilon,0}(\pm\infty) = \pm 1$ and $\varphi_{\varepsilon,1} = \varphi_{\varepsilon,0}^2 - 1$. We want $\varphi_{\varepsilon,0}(\lambda)^2 \approx 1$ and $\varphi_{\varepsilon,1}(\lambda) \approx 0$.

Choose a smooth nonzero bump function $\rho \in C_c^\infty(\mathbb{R})$ such that $\rho \geq 0$, ρ is even, $\int_{\mathbb{R}} \rho(\lambda) d\lambda = 1$, and $\text{supp}(\rho) \subseteq [-1/(6m), 1/(6m)]$. Let $\check{\rho}$ be the inverse Fourier transform. Then $\check{\rho}(0) = 1$ and $|\check{\rho}(\lambda)| < 1$ for $\lambda \neq 0$.

For a large even integer ε , define

$$C_\varepsilon = 2 \left(\int_{\mathbb{R}} \check{\rho}(\lambda)^\varepsilon d\lambda \right)^{-1}, \quad \varphi'_{\varepsilon,0}(\lambda) = C_\varepsilon \check{\rho}(\lambda)^\varepsilon.$$

Set $\varphi_{\varepsilon,0}(\lambda) = -1 + \int_{-\infty}^\lambda \varphi'_{\varepsilon,0}(t) dt$. Then $\int_{\mathbb{R}} \varphi'_{\varepsilon,0}(\lambda) d\lambda = 2$. Moreover,

$$\widehat{\varphi'_{\varepsilon,0}} = C_\varepsilon \underbrace{\rho * \cdots * \rho}_{\varepsilon \text{ factors}},$$

so

$$\text{supp}(\widehat{\varphi'_{\varepsilon,0}}) \subseteq \left[-\frac{\varepsilon}{6m}, \frac{\varepsilon}{6m} \right], \quad \text{supp}(\widehat{\varphi_{\varepsilon,1}}) \subseteq \left[-\frac{\varepsilon}{3m}, \frac{\varepsilon}{3m} \right].$$

Define the difference idempotent [MY26, (3.8)]

$$I_\varepsilon = \begin{pmatrix} \varphi_{\varepsilon,1}^+(\widetilde{D})^2 & -\varphi_{\varepsilon,1}^+(\widetilde{D})(1 - \varphi_{\varepsilon,1}^+(\widetilde{D}))\varphi_{\varepsilon,0}^-(\widetilde{D}) \\ -\varphi_{\varepsilon,1}^-(\widetilde{D})\varphi_{\varepsilon,0}^+(\widetilde{D}) & -\varphi_{\varepsilon,1}^-(\widetilde{D})^2 \end{pmatrix}.$$

Here $I_\varepsilon = P_\varepsilon - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, where P_ε is an idempotent.

Small propagation. We have $\text{prop}(I_\varepsilon) \leq \varepsilon/m$. Equivalently, $I_\varepsilon(x, y) = 0$ if $d(x, y) > \varepsilon/m$ [MY26, (3.9)].

Quantitative higher index [MY26, (3.13)].

$$[M] = \sum_{\sigma \in S_{m+1}} \text{sgn}(\sigma) \text{Tr}(I_\varepsilon(x_{\sigma(0)}, x_{\sigma(1)}) \cdots I_\varepsilon(x_{\sigma(m)}, x_{\sigma(0)})) [x_0, \dots, x_m].$$

The product is cyclic, the sum is antisymmetrized, and $m = \dim M$. The right-hand side is a cycle in the ε -homology group $H_m^\varepsilon(\widetilde{M}, \Gamma)$.

Definition 3.11 (ε -homology [MY26, (2.13)–(2.19)]). For $(x_0, \dots, x_k) \in \widetilde{M}^{k+1}$, define $\text{diam}(x_0, \dots, x_k) = \max_{i,j} d_{\widetilde{M}}(x_i, x_j)$. The ε -diagonal is

$$\text{diag}_\varepsilon(\widetilde{M}^{k+1}) = \{(x_0, \dots, x_k) : \text{diam}(x_0, \dots, x_k) \leq \varepsilon\}.$$

The chain group $C_k^\varepsilon(\widetilde{M}, \Gamma)$ consists of Γ -invariant measures supported in this set, where $\gamma(x_0, \dots, x_k) = (\gamma x_0, \dots, \gamma x_k)$. The boundary is

$$\partial \delta_{(x_0, \dots, x_k)} = \sum_{j=0}^k (-1)^j \delta_{(x_0, \dots, \widehat{x_j}, \dots, x_k)}, \quad \partial^2 = 0.$$

Thus the ε -homology group is $H_k^\varepsilon(\widetilde{M}, \Gamma) = H_k(C_*^\varepsilon(\widetilde{M}, \Gamma), \partial)$.

Intuition: it is generated by simplices of size at most ε .

3.5 The vanishing estimate

When ε is small, $H_k^\varepsilon(\widetilde{M}, \Gamma) \cong H_k(M)$. Choose an admissible rate $d > \psi_\Gamma$. Apply Schatten–Hölder and group decay to $[M]$. For sufficiently large i ,

$$\|M\| \leq C e^{d\varepsilon} \left\| (1 + \widetilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\|,$$

where the last norm is the operator norm on $L^2(\widetilde{M}, S_{\widetilde{M}})$, and $e^{d\varepsilon}$ comes from the exponential decay property of the fundamental group.

Assume $\text{Scal} \geq C > 0$. By Lichnerowicz,

$$\widetilde{D}^2 = \nabla^* \nabla + \frac{\text{Scal}}{4} \geq \frac{C}{4}.$$

Hence

$$\text{spec}(\widetilde{D}) \subseteq \left(-\infty, -\frac{\sqrt{C}}{2} \right] \cup \left[\frac{\sqrt{C}}{2}, \infty \right),$$

$$\text{Equivalently, } \text{spec}(\widetilde{D}) \cap \left(-\frac{\sqrt{C}}{2}, \frac{\sqrt{C}}{2} \right) = \emptyset.$$

Thus there is a spectral gap around 0.

We have $\varphi_{\varepsilon,1}(\widetilde{D}) = (\varphi_{\varepsilon,0}^2 - 1)(\widetilde{D})$. Choose $\varphi_{\varepsilon,0}$ so that $\varphi_{\varepsilon,0}^2$ is exponentially close to 1 for $|\lambda| \geq \sqrt{C}/2$.

Let $a = \sqrt{C}/2 > 0$ and $q = \sup_{|\lambda| \geq a} |\check{\rho}(\lambda)| < 1$. Then $|\check{\rho}(\lambda)|^\varepsilon \leq q^\varepsilon$ for $|\lambda| \geq a$, and

$$\sup_{|\lambda| \geq a} |\varphi_{\varepsilon,1}(\lambda)| \leq A \sqrt{\varepsilon} q^{\varepsilon-2}.$$

Let $q = e^{-b}$, where $b > 0$. Then

$$\sqrt{\varepsilon} q^{\varepsilon-2} = q^{-2} \sqrt{\varepsilon} e^{-b\varepsilon} \leq C e^{-b'\varepsilon}, \quad 0 < b' < b,$$

for sufficiently large ε . Thus

$$|\varphi_{\varepsilon,1}(\lambda)| \leq C e^{-c_m \varepsilon} \quad \text{for all } |\lambda| \geq \frac{\sqrt{C}}{2},$$

where c_m is a universal positive constant depending only on m .

The desired estimate is

$$\left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| \leq C' e^{-c_m \varepsilon}.$$

Indeed,

$$\begin{aligned} \left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| &\leq \sup_{|\lambda| \geq a} (1 + \lambda^2)^i \max \left\{ |\varphi_{\varepsilon,1}(\lambda)|^4, \right. \\ &\quad \left. |\varphi_{\varepsilon,1}(\lambda)(1 - \varphi_{\varepsilon,1}(\lambda))\varphi_{\varepsilon,0}(\lambda)|^2, |\varphi_{\varepsilon,1}(\lambda)\varphi_{\varepsilon,0}(\lambda)|^2 \right\} \\ &\leq C' \sup_{|\lambda| \geq a} (1 + \lambda^2)^i |\varphi_{\varepsilon,1}(\lambda)|^2 \\ &\leq C_i \varepsilon q^{2\varepsilon - 2N_i}. \end{aligned}$$

Here $N_i > 0$ depends only on i . Because

$$\begin{aligned} \|M\| &\leq C e^{d\varepsilon} \left\| (1 + \tilde{D}^2)^i I_\varepsilon^* I_\varepsilon \right\| \\ &\leq C e^{d\varepsilon} C' e^{-c_m \varepsilon} \\ &= C C' e^{(d - c_m)\varepsilon} \longrightarrow 0 \end{aligned}$$

as $\varepsilon \rightarrow \infty$, provided $d - c_m < 0$, we conclude that $\|M\| = 0$.

4 Higher index proof that the torus admits no positive scalar curvature

Let M be a complete Riemannian spin manifold, S the spinor bundle, and D the Dirac operator. The space $L^2(M; S)$ is the completion of $C_c^\infty(M; S)$ for the inner product $\langle u, v \rangle = \int_M \langle u(x), v(x) \rangle_x dx$, and

$$D : C_c^\infty(M; S) \subseteq L^2(M; S) \longrightarrow L^2(M; S).$$

Corollary 4.1 (Main result; [WY20, Corollary 3.3.5]). *The d -torus does not admit a metric of positive scalar curvature.*

For $u, v \in C_c^\infty(M; S)$, $\langle Du, v \rangle = \langle u, Dv \rangle$: D is formally self-adjoint. Also, $D^2 = \nabla^* \nabla + \kappa/4$, where ∇ is the spinor connection and κ is the scalar curvature [WY20, Theorem 3.1.5].

The graph is

$$\text{graph}(D) := \{(u, Du) : u \in C_c^\infty(M; S)\} \subseteq L^2(M; S) \oplus L^2(M; S).$$

Formal self-adjointness implies that D is closable; write $\overline{D}$ for its closure [WY20, Definitions D.1.2–D.1.3].

Definition 4.2 ([WY20, Definition D.1.4; Definition 8.1.6]). The minimal domain is $\mathcal{D}_{\min} = \text{Dom}(\overline{D})$. The maximal domain $\mathcal{D}_{\max}$ is the collection of all $v \in L^2(M; S)$ such that there exists $w \in L^2(M; S)$ such that, for all $u \in C_c^\infty(M; S)$, $\langle Du, v \rangle = \langle u, w \rangle$. The operator D is essentially self-adjoint if $\mathcal{D}_{\min} = \mathcal{D}_{\max}$.

A formally self-adjoint operator D is essentially self-adjoint if, whenever $v, w \in L^2(M; S)$ satisfy $\langle Du, v \rangle = \langle u, w \rangle$ for all $u \in C_c^\infty(M; S)$, there exists a sequence (v_n) in $C_c^\infty(M; S)$ such that $v_n \rightarrow v$ and $Dv_n \rightarrow w$ in L^2 -norm.

In local coordinates, if $D = \sum_j a_j(x) \frac{\partial}{\partial x_j} + b(x)$, then its symbol is $\sigma_D(x, \xi) = \sum_j \xi_j a_j(x)$ [WY20, Definition 8.1.3]. For the spin Dirac operator,

$$\sigma_D(x, \xi)^* = -\sigma_D(x, \xi), \quad \sigma_D(x, \xi)^2 = -\|\xi\|^2 1,$$

so $\sigma_D(x, \xi)^* \sigma_D(x, \xi) = \|\xi\|^2 1$. Thus, by [WY20, Definition 8.1.7],

$$c_D = \sup_{\substack{x \in M, \xi \in T_x^* M \\ \|\xi\|=1}} \|\sigma_D(x, \xi)\| = 1.$$

Proposition 4.3 ([WY20, Proposition 8.1.8]). *Let D be a formally self-adjoint first order differential operator. Assume that either the support of D is compact, or M is complete and c_D is finite. Then D is essentially self-adjoint.*

Consequently, for the spin Dirac operator:

Proposition 4.4 ([WY20, Proposition 3.2.1]). *Assume that the Riemannian metric on M is complete. Then D is essentially self-adjoint.*

In what follows, D denotes its self-adjoint closure when applying functional calculus or considering its spectrum and resolvent [WY20, Remark D.1.8]. Then, for any continuous bounded function $\phi : \mathbb{R} \rightarrow \mathbb{C}$, the functional calculus lets us define a bounded normal operator $\phi(D)$ on $L^2(M; S)$ [WY20, Theorem D.1.7].

Proposition 4.5 ([WY20, Proposition 3.2.4]).

- (1) $\text{supp } \widehat{\phi} \subseteq [-r, r] \implies \text{prop}(\phi(D)) \leq r$, where $\widehat{\phi}$ is the Fourier transform of ϕ .
- (2) If ϕ is compactly supported, then $\phi(D)$ is locally compact.

Definition 4.6 ([WY20, Definition 3.2.6]). The Roe algebra $C^*(M)$ of $L^2(M; S)$ is the C^* -algebra generated by all locally compact, finite propagation operators on $L^2(M; S)$.

If $\phi \in C_0(\mathbb{R})$, then $\phi(D) \in C^*(M)$. Let $\psi: \mathbb{R} \rightarrow [-1, 1]$ be continuous and odd, with $\lim_{x \rightarrow \pm\infty} \psi(x) = \pm 1$. Set $F = \psi(D)$. Then $\text{Ind}[F] \in K_i(C^*(M))$, where $i \equiv \dim M \pmod{2}$ [WY20, Lemma 3.2.7].

Theorem 4.7 ([WY20, Theorem 3.2.8]). If M is a complete spin Riemannian manifold with scalar curvature bounded below by a positive constant, then $\text{Ind}[F] = 0$.

Proof. For $u \in C_c^\infty(M; S)$,

$$\begin{aligned} \|Du\|^2 &= \langle D^2u, u \rangle \\ &= \|\nabla u\|^2 + \frac{1}{4} \int_M \kappa(x) |u(x)|^2 dx \\ &\geq \frac{1}{4} (\inf_M \kappa) \|u\|^2. \end{aligned}$$

Thus $\|Du\| \geq \frac{1}{2} \sqrt{\inf_M \kappa} \|u\|$ for $u \in \text{Dom}(\overline{D})$. For real λ with $|\lambda| < \frac{1}{2} \sqrt{\inf_M \kappa}$,

$$\begin{aligned} \|(D - \lambda)u\| &\geq \|Du\| - |\lambda| \|u\| \\ &\geq \left(\frac{1}{2} \sqrt{\inf_M \kappa} - |\lambda| \right) \|u\|. \end{aligned}$$

Hence $D - \lambda$ is injective and has closed range. Since D is self-adjoint and λ is real, $\text{Range}(D - \lambda)^\perp = \ker(D - \lambda) = \{0\}$. Therefore $\text{Range}(D - \lambda) = L^2(M; S)$, and

$$\|(D - \lambda)^{-1}\| \leq \left(\frac{1}{2} \sqrt{\inf_M \kappa} - |\lambda| \right)^{-1}.$$

Choose $0 < c < \frac{1}{2} \sqrt{\inf_M \kappa}$. Then $\text{spec}(D) \cap [-c, c] = \emptyset$. Choose ψ as above with $\psi^2 = 1$ on the spectrum of D , without changing the index [WY20, Lemma 3.2.7]. Set $F = \psi(D)$. Then $F^2 = (\psi^2)(D) = 1$, so $\text{Ind}[F] = 0$ in $K_i(C^*(M))$ [WY20, Lemma 2.8.7]. $\square$

Proof of Corollary 4.1. Apply the construction to the universal cover of the torus. Let $M = (S^1)^d$ carry an arbitrary Riemannian metric and spin structure. Let $\tilde{M}$ be the universal cover with the complete lifted metric, $\tilde{S}$ its spinor bundle, and $\tilde{D}$ its Dirac operator. Let $\chi: \mathbb{R} \rightarrow [-1, 1]$ be continuous and odd, with $\lim_{x \rightarrow \pm\infty} \chi(x) = \pm 1$. Set $\tilde{F} = \chi(\tilde{D})$; then $\text{Ind}[\tilde{F}] \in K_i(C^*(\tilde{M}))$. For $t \in [1, \infty)$, let $\tilde{F}_t := \chi(t^{-1}\tilde{D})$. The family $(\tilde{F}_t)$ is a multiplier of the localisation algebra $C_L^*(\tilde{M})$ [WY20, Definition 3.3.1], such that $(1 - \tilde{F}_t^2) \in C_L^*(\tilde{M})$. Thus $\text{Index}_L(\tilde{F}) \in K_i(C_L^*(\tilde{M}))$ [WY20, Lemma 3.3.2].

The group $K_i(C_L^*(\tilde{M})) \cong \mathbb{Z}$, and $\text{Index}_L(\tilde{F})$ is a generator [WY20, Proposition 3.3.3]. Hence $\text{Index}_L(\tilde{F}) \neq 0$. Evaluation at one is the $*$ -homomorphism

$$\begin{aligned} \text{ev}: C_L^*(\tilde{M}) &\longrightarrow C^*(\tilde{M}), \\ (T_t) &\longmapsto T_1. \end{aligned}$$

It satisfies

$$\text{ev}_*(\text{Index}_L(\tilde{F})) = \text{Ind}[\tilde{F}].$$

Since ev_* is injective [WY20, Theorem 3.3.4], $\text{Ind}[\tilde{F}] \neq 0$. If M had positive scalar curvature, its compactness would give $\inf_M \kappa > 0$, and the complete lifted metric would have the same lower bound. Theorem 4.7 would give $\text{Ind}[\tilde{F}] = 0$, a contradiction. So the d -torus admits no metric of positive scalar curvature. $\square$

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